University of Cambridge Mathematical Tripos Part II

Analysis of Functions

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Course schedule

Lebesgue integration theory
Review of integration: simple functions, monotone and dominated convergence; existence of Lebesgue measure; definition of L^p spaces and their completeness. The Lebesgue differentiation theorem. Egorov's theorem, Lusin's theorem. Mollification by convolution, continuity of translation and separability of L^p when $p \neq \infty$. [5]

Banach and Hilbert space analysis
Strong, weak and weak-* topologies; reflexive spaces. Review of the Riesz representation theorem for Hilbert spaces; the Radon–Nikodym theorem; the dual of L^p. Compactness: review of the Ascoli–Arzelà theorem; weak-* compactnesss of the unit ball for separable Banach spaces. The Riesz representation theorem for spaces of continuous functions. The Hahn–Banach theorem and its consequences: separation theorems; Mazur's theorem. [7]

Fourier analysis
Definition of Fourier transform in L^1; the Riemann–Lebesgue lemma. Fourier inversion theorem. Extension to L^2 by density and Plancherel's isometry. Duality between regularity in real variable and decay in Fourier variable. [3]

Generalized derivatives and function spaces
Definition of generalized derivatives and of the basic spaces in the theory of distributions: $\mathcal{D}/\mathcal{D}'$ and $\mathcal{S}/\mathcal{S}'$. The Fourier transform on $\mathcal{S}'$. Periodic distributions; Fourier series; the Poisson summation formula. Definition of the Sobolev spaces H^s in $\mathbb{R}^d$. Sobolev embedding. The Rellich–Kondrashov theorem. The trace theorem. [5]

Applications
Construction and regularity of solutions for elliptic PDEs with constant coefficients on $\mathbb{R}^n$. Construction and regularity of solutions for the Dirichlet problem of Laplace's equation. The spectral theorem for the Laplacian on a bounded domain. *The direct method of the Calculus of Variations.* [4]

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1 Measure theory

Recap of elementary measure theory

We recall some basic notions from II Probability and Measure whilst establishing the notation for this course.

Let E be a set. A family $\mathcal{B} \subset \mathcal{P}(E)$ is a σ-algebra if $\varnothing \in \mathcal{B}$ and it is closed under countable unions and complements. A map $\mu\colon \mathcal{B} \to [0,\infty]$ is a measure if it is σ-additive, i.e., $\mu(\bigsqcup_n A_n) = \sum_n \mu(A_n)$ for a countable disjoint collection (A_n). A pair $(E,\mathcal{B})$ is called a measurable space, whereas a triple $(E,\mathcal{B},\mu)$ is called a measure space. If E is a topological space, we can (and, in this course, will) consider $\mathcal{B}$ to be the Borel σ-algebra.

A map $f\colon E \to \mathbb{C}$ is measurable if $f^{-1}(A) \in \mathcal{B}$ for all Borel $A \subset \mathbb{C}$. If $f\colon E \to [0,\infty]$ is measurable, then $\int f\,d\mu$ is well-defined (in $[0,\infty]$). We say that $f\colon E \to \mathbb{C}$ is integrable if it is measurable and $\int |f|\,d\mu < \infty$. We then denote by $\mathcal{L}^1(E,\mathcal{B},\mu)$ the set of all integrable functions $E \to \mathbb{C}$. Writing $f \sim g \iff f = g$ a.e., we then define $L^1(E,\mathcal{B},\mu) := \mathcal{L}^1(E,\mathcal{B},\mu)/\sim$. However, it is of course standard to refer to an equivalence class $[f] \in L^1(E)$ using a concrete function f from it.

Theorem 1.1 Dominated convergence theorem
Let $(E,\mathcal{B},\mu)$ be a measure space. Let $g, f, f_1, f_2 \in L^1(E,\mathcal{B},\mu)$. Suppose $f_n(x) \to f(x)$ and $|f_n(x)| \le g(x)$ for a.e. $x \in E$. Then

$$\int_E f_n\,d\mu \to \int_E f\,d\mu$$

Recall that a measure space $(E,\mathcal{B},\mu)$ is σ-finite if there exists $(A_n)_{n\in\mathbb{N}}$ such that $E = \bigcup_{n\in\mathbb{N}} A_n$ and $\mu(A_n) < \infty$ for all $n \in \mathbb{N}$.

Theorem 1.2 Fubini's theorem
Let $(E,\mathcal{A},\mu)$ and $(F,\mathcal{B},\mu)$ be two σ-finite measure spaces. Let $f\colon E\times F \to \mathbb{C}$ be $\mathcal{A}\otimes\mathcal{B}$-measurable.

(i) *If the map $x \mapsto \int_F f(x,y)\,d\nu$ in $L^1(E,\mathcal{A},\mu)$, then $f \in L^1(E\times F)$.*

(ii) *If $f \in L^1(E\times F)$, then*

$$\int_{E\times F} f\,d\mu\otimes\nu = \int_E\int_F f(x,y)\,d\nu(y)\,d\mu(x) = \int_F\int_E f(x,y)\,d\mu(x)\,d\nu(y)$$

1.1 Signed and complex measures

Definition Complex and signed measures
Let $(E,\mathcal{B})$ be a measurable space. A set function $\mu\colon \mathcal{B} \to \mathbb{C}$ is a complex measure if it is σ-additive. We then say that $(E,\mathcal{B},\mu)$ is a complex measure space. If $\mu(\mathcal{B}) \subset \mathbb{R}$, we call μ a signed measure and $(E,\mathcal{B},\mu)$ a signed measure space.

Remark. Previously, we really have been considering *positive* measures, though we will always just refer to such maps as *measures*. Note that not every (positive) measure is a complex measure.

Given a complex measure μ, define the *real* and *imaginary parts* $\operatorname{Re}\mu$ and $\operatorname{Im}\mu$ as

$$\forall\, A \in \mathcal{B} \qquad \operatorname{Re}\mu(A) = \operatorname{Re}(\mu(A)), \qquad \operatorname{Im}\mu(A) = \operatorname{Im}(\mu(A))$$

It is easy to verify that these are signed measures.

Definition Positive and negative sets
Let $(E, \mathcal{B}, \mu)$ be a signed measure space. We say that A is a positive (resp. negative) set if $\mu(B) \geq 0$ (resp. $\mu(B) \leq 0$) for all $B \subset A$.

Theorem 1.3 Hahn decomposition
Let $(E, \mathcal{B}, \mu)$ be a signed measure space. Then there exists a positive set $P \in \mathcal{B}$ and a negative set $N \in \mathcal{B}$ such that $E = P \sqcup N$.

In proving the Hahn decomposition theorem, we will ultimately be defining P be a positive set of the largest possible measure. However, we first show the existence of non-trivial positive sets via the following lemma.

Lemma 1.4
For all $A \subset \mathcal{B}$, there is a positive set $D \subset A$ such that $\mu(D) \geq \mu(A)$.

Proof. If A is positive, then we can simply take $D = A$ and we are done. It remains to carefully consider the case A is not positive. Pick negative set $B_1 \subset A$ such that $\mu(B_1)$ is "as negative as possible" in the approximate sense that there is no $B \subset A$ and $k \in \mathbb{N}$ for which $\mu(B_1) > -1/k \geq \mu(B)$. Let $A_1 = A \backslash B_1$. Continue inductively to define $B_2, A_2, B_3, A_3, \cdots$ in such a way that $B_{j+1} \subset A_j$ and $\mu(B_{j+1}) > -1/k$ for some $k \in \mathbb{N}$ only if this is so for all subsets of A_i; we then take $A_{j+1} = A_j \backslash B_{j+1}$.

Now, take $D = \bigcap A_j$. Then $A = D \sqcup B_1 \sqcup B_2 \sqcup \cdots$, so $\mu(A) = \mu(D) + \sum_j \mu(B_j)$. Since $\mu(B_j) \geq 0$ for each j, we have $\mu(D) \geq \mu(A)$. It then remains to show that D is positive. Note that $\mu(B_j) \to 0$ by convergence of $\sum_j \mu(B_j)$. Fix $k \in \mathbb{N}$. Then $\mu(B_i) > -1/k$ for some i. Then $\mu(B) > -1/k$ for all $B \subset A_{i-1}$ and thus for all $B \subset D$. Hence, D must be positive. ■

Proof of Theorem 1.3. Note that a union of positive sets $A = A_1 \cup A_2 \cup \cdots$ is positive; indeed, for any $B \subset A$, we have $B = (B \cap A_1) \sqcup (B \cap A_2 \backslash A_1) \sqcup (B \cap A_3 \backslash (A_1 \cup A_2)) \sqcup \cdots$ which implies that $\mu(B) \geq 0$ by σ-additivity.

Let $s = \sup\{\mu(A) \colon A \subset E \text{ positive}\}$ and let (P_i) be a sequence of positive sets in $\mathcal{B}$ such that $\lim \mu(P_i) = s$. Then by above, we know that $P = \bigcup P_i$ is positive. Since $\mu(P) \geq \mu(P_i)$, we also have $\mu(P) = s$.

Now, suppose, on the contrary, that $N := E \backslash P$ is not a negative set. Then there exists $B \subset N$ such that $\mu(B) > 0$. By Lemma 1.4, there exists $D \subset B$ positive such that $\mu(D) \geq \mu(B)$. It then follows that $\mu(P \cup D) > \mu(P)$ with $P \cup D$ a positive set — a contradiction! ■

Remarks

- A key corollary of the Hahn decomposition theorem is the Jordan decomposition of signed measures.
- The decomposition $E = P \sqcup N$ is manifestly non-unique as we can, for instance, 'move' a negligible subset from P to N. However, this makes no difference in the Jordan decomposition of μ.

Corollary 1.5 Jordan decomposition of signed measures
Let $(E, \mathcal{B}, \mu)$ be a signed measure space, with $E = P \sqcup N$ for positive P and negative N. Then $\mu^+ := \mu|_P$ and $\mu^- := -\mu|_N$ are positive measures satisfying $\mu = \mu^+ - \mu^-$.

Proof. Trivial. ■

Now that we've proven the Hahn and Jordan decompositions, we can now extend Lebesgue integration to signed and complex measures.

Definition Integral with respect to signed measures
Let $(E, \mathcal{B}, \mu)$ be a signed measure space. A measurable function $f\colon E \to \mathbb{C}$ is integrable if it is integrable with respect to μ^+ and μ^-. In that case, we then define its integral to be

$$\int_E f\, d\mu := \int_E f\, d\mu^+ - \int_E f\, d\mu^-$$

Definition Integral with respect to complex measures
Let $(E, \mathcal{B}, \mu)$ be a complex measure space. A measurable function $f\colon E \to \mathbb{C}$ is integrable with respect to $\operatorname{Re}\mu$ and $\operatorname{Im}\mu$. In that case, we then define its integral to be

$$\int_E f\, d\mu := \int_E f\, d\operatorname{Re}\mu + i\int_E f\, d\operatorname{Im}\mu$$

Finally, we end this subsection with a brief discussion on the Banach space structure of the space of complex measures. Given a signed measure μ, we can define its *total variation measure* to be

$$|\mu| := \mu^+ + \mu^-$$

and its *total variation norm* to be

$$||\mu|| := |\mu|(E) = \mu^+(E) + \mu^-(E)$$

On the Example Sheet, you will extend these notions to complex measures. It can then be shown that the space of complex measures on a measurable space $(E, \mathcal{B})$ forms a Banach space with respect to the total variation norm.

1.2 Radon-Nikodym theorem

Motivation: In II Probability and Measure, you met some examples of continuous and discrete random variables. Are there probability distributions that do not belong to either category? How do we know if a random variable has a density?

Let us start off with some definitions. Let μ, ν be (positive) measures on $(E, \mathcal{B})$.

Definition Absolutely continuous measure
We say that ν is absolutely continuous wrt μ if, for all $A \in \mathcal{B}$, $\mu(A) = 0 \implies \nu(A) = 0$. We then write $\nu \ll \mu$.

Definition Singular measure
We say that ν is singular wrt μ if there exists a decomposition $E = A \sqcup B$ such that $\nu(A) = 0$ and $\mu(B) = 0$. We then write $\nu \perp \mu$.

Remark. It is easy to see that $\nu \perp \mu \iff \mu \perp \nu$.

Definition Concentrated on a set
We say that μ is concentrated on a set $A \in \mathcal{B}$ if $\mu(E \backslash A) = 0$.

Remark. $\mu \perp \nu$ iff there exist disjoint sets $A, B \in \mathcal{B}$ such that μ is concentrated on A and ν is concentrated on B.

We now state the two main theorems of this subsection.

Theorem 1.6 Radon-Nikodym theorem
Let μ, ν be finite measures on $(E, \mathcal{B})$. Suppose that $\nu \ll \mu$. Then there exists $f \in L^1(E, \mathcal{B}, \mu)$ such that

$$\nu(A) = \int_A f \, d\mu$$

for all $A \in \mathcal{B}$. If g is another function that satisfies the conclusion of this theorem, then $f = g$ μ-a.e.

The function f in the theorem above is called the *Radon-Nikodym (RN) derivative*, often denoted by

$$\frac{d\nu}{d\mu}$$

Remark. We could extend this theorem by replace finite with σ-finite, but this comes at the expense of having $f \in L^1$.

Theorem 1.7 Lebesgue decomposition
Let μ, ν be finite measures. Then there exist unique measures ν_a, ν_s such that $\nu = \nu_a + \nu_s$ with $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

The idea behind the proof of these theorems is as follows. For each $t \in \mathbb{R}_{\geq 0}$, let $P_t \sqcup N_t$ be a Hahn-decomposition of $\nu - t\mu$. Then, for all measurable $A \subset P_t$, we have $(\nu - t\mu)(A) \geq 0$ and so $\nu(A) \geq t\mu(A)$. Similarly, for all measurable $A \subset N_t$, we have $\nu(A) \leq t\mu(A)$. Let $t_1 < t_2$. if $A \in P_{t_1} \cap N_{t_2}$, then

$$t_1\mu(A) \leq \nu(A) \leq t_2\mu(A)$$

so it is reasonable to expect that the Radon-Nikodym derivative will be between t_1 and t_2 on $P_{t_1} \cap N_{t_2}$.

The above intuition motivates the following definition: For each $n \in \mathbb{Z}_{\geq 0}$, define

$$f_n(x) := \sup\{t \in 2^{-n}\mathbb{Z}_{\geq 0} \colon x \in P_t\}$$

where $2^{-n}\mathbb{Z}_{\geq 0}$ is the set of all numbers of the form $2^{-n}a$ with $a \in \mathbb{Z}_{\geq 0}$. Set $f(x) := \lim_{n\to\infty} f_n(x)$ for each x.

Lemma 1.8
There exists $A \in \mathcal{B}$ such that $\mu(E \backslash A) = 0$ and

$$\nu(B) = \int_B f \, d\mu$$

for all $B \subset A$.

Proof. For each $n \in \mathbb{Z}_{\geq 0}$, let $A_n := \{x \colon f_n(x) < \infty\}$. We first show that $\mu(E \backslash A_n) = 0$. To do this, fix $t \in 2^{-n}\mathbb{Z}_{\geq 0}$. Then $E \backslash A_n \subset \bigcup_{s \geq t} P_s$, with $\bigcup_{s \geq t} P_s$ positive for $\nu - t\mu$. We thus deduce that $\nu(E) \geq \nu(E \backslash A_n) \geq t\mu(E \backslash A_n)$. Taking $t \to \infty$, we conclude that $\mu(E \backslash A_n) = 0$.

Now, fix $B \subset A_n$. For $s \in 2^{-n}\mathbb{Z}_{\geq 0}$, let

$$B_t = \{x \in B \colon f_n(x) = t\}$$

Note that $B = \bigsqcup_t B_t$ and that, if $x \in B_t$, then $x \in P_t$ but $x \notin P_{t+2^{-n}}$. Thus, $B_t \subset P_t \cap N_{t+2^{-n}}$ and so we have

$$t\mu(B_t) \leq \nu(B_t) \leq (t + 2^{-n})\mu(B_t)$$

This implies that

$$\left|\nu(B_t) - \int_{B_t} f_n \, d\mu\right| \leq 2^{-n}\mu(B_t)$$

from which it follows that

$$\left|\nu(B) - \int_{B} f_n \, d\mu\right| \leq 2^{-n}\mu(B)$$

Finally, set $A := \bigcap_n A_n$. Picking $B \subset A$, the above holds for all n, so by the monotone convergence theorem, we conclude that

$$\int_B f \, d\mu = \lim_{n\to\infty} \int_B f_n \, d\mu = \nu(B)$$

Note also that $\mu(E\backslash A) = \mu(\bigcup(E\backslash A_n)) = 0$. ■

Proof of Theorem 1.6. Since $\nu \ll \mu$, we have $\nu(E\backslash A) = 0$. Then for each $B \in \mathcal{B}$,

$$\nu(B) = \nu(B \cap A) + \nu(B\backslash A) = \int_{B\cap A} f \, d\mu = \int_B f \, d\mu$$

Thus, f is a Radon-Nikodym derivative.

It remains to show uniqueness. Let f, g be two RN derivatives. Suppose that, on the contrary, $f \neq g$ μ-a.e. Then there exists $\varepsilon > 0$ such that either $B = \{x\colon f(x) - g(x) > \varepsilon\}$ or $\{x\colon f(x) - g(x) < \varepsilon\}$ has positive measure. WLOG, let it be the first one. Then we have

$$0 = \nu(B) - \nu(B) = \int_B (f - g) \, d\mu \geq \int_B \varepsilon \, d\mu = \varepsilon\mu(B)$$

as required. ■

Proof of Theorem 1.7. We first prove existence. Pick $A \in \mathcal{B}$ as in the previous lemma. Set $\nu_a = \nu|_A$ and $\nu_s = \nu|_{E\backslash A}$. Note that ν_s is concentrated on $E\backslash A$ whereas μ is concentrated on A, so $\nu \perp \mu$. Moreover, $\nu_a \ll \mu$ since f is a RN derivative for ν_a.

For uniqueness, suppose $\nu = \nu_a + \nu_s = \tilde{\nu}_a + \tilde{\nu}_s$. Then we can pick μ-null sets $D, \tilde{D}$ on which $\nu_s, \tilde{\nu}_s$ concentrate. Then $F := D \cup \tilde{D}$ is a μ-null set on which both ν_s and $\tilde{\nu}_s$ concentrate. Now, fix $G \in \mathcal{B}$. Since $\nu_a, \tilde{\nu}_a \ll \mu$, we have

$$\nu_a(G \cap F) = \tilde{\nu}_a(G \cap F) = 0$$

On the other hand, since $\nu_s, \tilde{\nu}_s$ both concentrate on F, we have

$$\nu_s(G\backslash F) = \tilde{\nu}_s(G\backslash F) = 0$$

It then follows that

$$\nu_a(G\backslash F) = \tilde{\nu}_a(G\backslash F) + \tilde{n}u_s(G\backslash F) - \nu_s(G\backslash F) = \tilde{\nu}_a(G\backslash F)$$

Putting everything together, we obtain $\nu_a(B) = \tilde{\nu}_a(B)$ and hence $\nu_a = \tilde{\nu}_a$. ■

Example *Bernoulli convolution*
Fix $\lambda \in (0, 1)$. Let $(X_n)_{n\in\mathbb{N}}$ be an iid sequence of random variables that take values in $\{0, 1\}$ with equal probabilities. Let ν_λ be the distribution of the random variable

$$Y = \sum_{n=1}^{\infty} \lambda^n X_n$$

The first m values $X_1, \cdots, X_m$ determine the value of Y up to an error of at most $\sum_{n=m+1}^{\infty} 3^{-n} = 2/3^{m+1}$. If K is the set of values of Y, then it can be covered by 2^m intervals of length $2/3^{m+1}$. Then we have the estimate

$$|K| \leq 2^m \cdot \frac{2}{3^{m+1}} \to 0$$

On the other hand, $\nu_{1/3}(\mathbb{R} \backslash K) = 0$, showing that $\nu_{1/3} \perp dx$.

Remark. Solomyak has proven that $\nu_\lambda \ll dx$ for almost all $\lambda \in [1/2, 1)$. However, in a later example sheet, you will see that $\nu_\varphi \perp dx$ for $\varphi = (\sqrt{5} - 1)/2$. It remains a major open problem whether $v_{2/3} \ll dx$ or $\nu_{2/3} \perp dx$.

1.3 Lebesgue differentiation theorem

Motivation: From the previous subsection we know that an absolutely continuous measure $\mu \ll dx$ has a Radon-Nikodym derivative. But how do we find it? One might try to consider

$$\frac{\mu(B(x,r))}{|B(x,r)|}$$

and try to show that it converges to the Radon-Nikodym derivative as $r \to 0$. Indeed, this can be done via the Lebesgue differentiation theorem.

Another reason for studying this theorem is to obtain a generalisation of the fundamental theorem of calculus for the theory of Lebesgue integration.

Definition Lebesgue point
Let $f \in L^1(\mathbb{R}^d)$. A point $x \in \mathbb{R}^d$ is a Lebesgue point of f if

$$\lim_{r \to 0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy = 0$$

Remark. If x is a Lebesgue point of f, then

$$\left| \frac{1}{|B(x,r)|} \int_{B(x,r)} f(y) \, dy - f(x) \right| \leq \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \to 0$$

which gives us a generalisation of $F'(x) = f(x)$ that also makes sense in higher dimensions.

Theorem 1.9 Lebesgue differentiation theorem
Let $f \in L^1(\mathbb{R}^d)$. Then Lebesgue-a.e. $x \in \mathbb{R}^d$ is a Lebesgue point of f.

Definition Hardy-Littlewood maximal function
The Hardy-Littlewood maximal function Mf of $f \in L^1(\mathbb{R}^d)$ is

$$Mf(x) = \sup_{r \in \mathbb{R}_{>0}} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(t)| \, dt$$

Proposition 1.10 Maximal inequality
Let $f \in L^1(\mathbb{R}^d)$ and $t \in \mathbb{R}_{>0}$. Then

$$|\{x \in \mathbb{R}^d \colon Mf(x) > t\}| \leq \frac{5^d}{t} ||f||_1$$

Remark. The constant 5^d can be substantially improved; the best constant is known to grow at most linearly in d. It remains an open problem whether or not the constant can be made independent of d.

Now, let us prove the Lebesgue differentiation theorem assuming the above proposition. To do this, we will make use of the following density result which we will prove in the subsequent subsection.

Lemma 1.11
For all $f \in L^1(\mathbb{R}^d)$ and $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^d)$ such that $||f - g||_1 < \varepsilon$.

Proof of Theorem 1.9. Fix $f \in L^1(\mathbb{R}^d)$. For each $t \in \mathbb{R}_{>0}$, define

$$A_t = \left\{ x \in \mathbb{R}^d \colon \limsup_{r \to 0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy > 2t \right\}$$

Observe that it suffices to show that $|A_t| = 0$ for all $t > 0$.

Fix $\varepsilon > 0$. By density of $C_c(\mathbb{R}^d)$ in $L^1(\mathbb{R}^d)$, we can pick $g \in C_c(\mathbb{R}^d)$ such that $||f - g||_1 < \varepsilon$. By the triangle inequality, we obtain the estimate

$$\begin{aligned} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \leq \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - g(y)| \, dy \\ + \frac{1}{|B(x,r)|} \int_{B(x,r)} |g(y) - g(x)| \, dy \\ + |f(x) - g(x)| \end{aligned}$$

Since g is continuous, we have

$$\limsup_{r \to 0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |g(y) - g(x)| \, dy = 0$$

so the estimate becomes

$$\frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y) - f(x)| \, dy \leq M[f - g](x) + |f(x) - g(x)|$$

Now, if $x \in A_t$, then either $|M[f-g](x)| > t/2$ or $|f(x) - g(x)| > t/2$. By the maximal inequality (Proposition 1.10), we can estimate

$$|\{x \in \mathbb{R}^d \colon M[f - g](x) > t\}| \leq \frac{5^d}{t} ||f - g||_1 < \frac{5^d \varepsilon}{t}$$

On the other hand, we can use Markov's inequality to estimate

$$|\{x \in \mathbb{R}^d \colon |f(x) - g(x)| > t\}| \leq \frac{||f - g||_1}{t} < \frac{\varepsilon}{t}$$

Putting everything together, we have, for each $t > 0$,

$$|A_t| < \frac{5^d + 1}{t} \varepsilon$$

Taking $\varepsilon \to 0$ yields the required result. ■

Now, we need to prove the maximal inequality. Our proof will make use of the following lemma:

Lemma 1.12 Vitali covering lemma
Let $\mathcal{U}$ be a collection of balls in $\mathbb{R}^d$ with bounded diameter. Then there is a disjoint (finite or countably infinite) collection $\{V_1, V_2, \cdots\} \subset \mathcal{U}$ such that

$$\left| \bigcup_j V_j \right| = \infty \qquad \text{or} \qquad \bigcup_{U \in \mathcal{U}} U \subset \bigcup_j 5 \cdot V_j$$

where $5 \cdot V_j$ denotes the dilation of the ball V_j about its centre by a factor of 5.

Proof. We will construct the collection inductively. Pick $V_1 \in \mathcal{U}$ such that $\operatorname{diam}(V_1) \geq \operatorname{diam}(U)/2$ for all $U \in \mathcal{U}$. Suppose we have chosen $V_1, \cdots, V_n$ for some $n \geq 1$. Then pick $V_{n+1} \in \mathcal{U}$ such that it is disjoint from $V_1 \cup \cdots \cup V_n$ and $\operatorname{diam}(V_{n+1}) \geq \operatorname{diam}(U)/2$ for all $U \in \mathcal{U}$ for which U is disjoint from $V_1 \cup \cdots \cup V_n$.

If $|\bigcup_j V_j| = \sum_j |V_j| = \infty$, then we are done. Otherwise, we must have $\{V_j\}_j$ is finite or $\operatorname{diam}(V_n) \to 0$. Fix $U \in \mathcal{U}$. In either case, U will not be disjoint from $V_1 \cup \cdots \cup V_n$ for some n. Let N be the smallest such n. This implies that $U \cap V_N \neq \varnothing$. By construction, we have $\operatorname{diam}(U) \leq 2\operatorname{diam}(V_N)$. As an exercise, you can check (using the triangle inequality) that $U \subset 5 \cdot V_N$, completing the proof. ■

Proof of Proposition 1.10. Note that, for each $x \in \mathbb{R}^d$ such that $Mf(x) > t$, we can pick $r(x) > 0$ such that

$$|B(x, r(x))| \leq t^{-1} \int_{B(x,r(x))} |f(t)|\, dt \leq t^{-1}||f||_1 \tag{†}$$

Denote by $U(x) = B(x, r(x))$. Observe that

$$\{x \in \mathbb{R}^d \colon Mf(x) > t\} \subset \bigcup_{x\colon Mf(x)>t} U(x)$$

We want

$$\left|\bigcup U(x)\right| \leq t^{-1} \int_{\bigcup U(x)} |f(t)|\, dt$$

Of course, if the balls are disjoint, the above does hold by σ-additivity. While this is not necessarily the case, the Vitali covering lemma helps! By (†), the set $\{\operatorname{diam}(U(x)) \colon x \in \mathbb{R}^d, Mf(x) > t\}$ is bounded, so we can apply the lemma to $\mathcal{U} = \{U(x) \colon Mf(x) > t\}$. Each V_j given by the lemma satisfies (†), so we have

$$\left|\bigcup_{x\colon Mf(x)>t} U(x)\right| \leq \left|\bigcup_j 5 \cdot V_j\right| \leq 5^d \sum_j |V_j| \leq 5^d t^{-1}||f||_1$$

as required. ■

1.4 Littlewood's principles

In his *Lectures on the Theory of Functions*, Littlewood stated three principles: "Every (measurable) set is nearly a finite sum of intervals; every function (of class L^p) is nearly continuous; every convergent sequence of functions is nearly uniformly convergent."

The first principle can be stated precisely as follows:

Proposition 1.13
Let $A \subset \mathbb{R}^d$ with $|A| < \infty$. Then for every $\varepsilon > 0$, there exists $B \subset \mathbb{R}^d$ which is a finite union of rectangles, such that $|A \triangle B| < \varepsilon$.

Exercise. Prove the above proposition.

Next, a more precise formulation of the second principle is given by Lusin's theorem. Its proof makes use of the regularity of finite Borel measures on metric spaces, which we now recall from II Probability and Measure.

Lemma 1.14 Regularity of finite Borel measures
Let (E, dist) be a metric space, $\mathcal{B}$ be the Borel σ-algebra of E, and μ a finite Borel measure on $(E, \mathcal{B})$. Then for all $A \in \mathcal{B}$ and $\varepsilon > 0$, there exists a closed set $K \subset E$ and an open set $U \subset E$ such that $K \subset A \subset U$ and $\mu(U \backslash K) < \varepsilon$.

Theorem 1.15 (Weak) Lusin's theorem
Let (E, dist) be a metric space and μ a finite Borel measure. If $f\colon E \to \mathbb{C}$ is measurable, then for every $\varepsilon > 0$, there exists a closed set $K \subset E$ such that $\mu(E \backslash K) < \varepsilon$ and $f|_K$ is continuous.

Proof. We will prove this result in steps: we first consider the special case of indicator functions and gradually extend the result to measurable functions.

Step 1: Indicator functions on Borel sets
Let $A \in \mathcal{B}$. Fix $\varepsilon > 0$. By regularity of μ, we can pick closed $F \subset E$ and open $U \subset E$ such that $F \subset A \subset U$ and $\mu(U \backslash F) < \varepsilon$. Recall from II Linear Analysis that metric spaces are normal, so by Urysohn's lemma, there exists a continuous function $g\colon E \to [0, 1]$ such that $g|_{E \backslash U} = 0$ and $g|_F = 1$. In this case, we can set $K = (E \backslash U) \cup F$. Note that $\mathbb{1}_A|_K = g|_K$ is continuous, and $\mu(E \backslash K) = \mu(U \backslash F) < \varepsilon$, as required.

Step 2: Simple functions
Fix $\varepsilon > 0$. Suppose f is a simple function, given by $f = \sum_{i=1}^n \alpha_i \mathbb{1}_{A_i}$ with each $A_i \in \mathcal{B}$. Apply Step 1 for each $\mathbb{1}_{A_i}$ to obtain closed sets K_i satisfying $\mu(E \backslash K_i) < \varepsilon/n$ and $\mathbb{1}_{A_i}|_{K_i}$ continuous. Set $K = \bigcap_{i=1}^n K_i$. Note that K is closed, $\mu(E \backslash K) \leq \sum_{i=1}^n \mu(E \backslash K_i) < \varepsilon$ and $\mathbb{1}_{A_i}|_K$ continuous.

Step 3: Bounded measurable functions
Fix $\varepsilon > 0$. Suppose f is a bounded measurable function. Recall from II Probability and Measure that simple functions are dense in $L^\infty(E)$, so we can pick a sequence $(f_n)_{n \in \mathbb{N}}$ of simple functions that uniformly converge to f. For each f_n, apply Step 2 to get closed K_n such that $\mu(E \backslash K_n) < \varepsilon 2^{-n}$ and $f_n|_{K_n}$ is continuous. Set $K = \bigcap_{n \in \mathbb{N}} K_n$. Note that K is closed, $\mu(E \backslash K) \leq \sum_{n=1}^\infty \mu(E \backslash K_n) < \varepsilon$ and $f|_K$ is continuous (by the uniform limit theorem).

Step 4: Measurable functions
Fix $\varepsilon > 0$. Suppose f is a measurable function. For each $n \in \mathbb{N}$, define $E_n = \{x \in E\colon |f(x)| \leq n\}$. Note that $(\mu(E_n))_{n \in \mathbb{N}}$ is a monotonically increasing sequence, bounded above by $\mu(E) < \infty$. Thus, $\mu(E_n) \to \mu(E)$. Pick $N \in \mathbb{N}$ such that $\mu(E \backslash E_N) < \varepsilon/2$. Apply Step 3 to the bounded measurable function $f \mathbb{1}_{E_N}$, yielding a closed $\tilde{K} \subset E$ such that $\mu(E \backslash \tilde{K}) < \varepsilon/2$ and $f \mathbb{1}_{E_N}|_{\tilde{K}}$ is continuous. Now, consider the Borel set $\tilde{K} \cap E_N$. Note that $\mu(E \backslash (\tilde{K} \cap E_N)) \leq \mu(E \backslash E_N) + \mu(E_N \backslash K) < \varepsilon$. By inner regularity of μ, we can pick closed $K \subset E$ such that $K \subset \tilde{K} \cap E_N$ and $\mu((\tilde{K} \cap E_N) \backslash K) < \varepsilon - \mu(E \backslash (\tilde{K} \cap E_N))$. Then $\mu(E \backslash K) < \varepsilon$ and $f|_K$ is continuous. ■

Theorem 1.16 (Strong) Lusin's theorem
Let (E, dist) be a metric space and μ a finite Borel measure. If $f\colon E \to \mathbb{C}$ is measurable, then, for every $\varepsilon > 0$, there exists a closed set $K \subset E$ and $g \in C(E)$ such that $f = g$ on K, and $\mu(E \backslash K) < \varepsilon$. If, in addition, f is also bounded, then the conclusion holds with $||g||_\infty = ||f||_\infty$.

Proof. We simply apply Tietze's extension theorem (cf. II Linear Analysis, including variants in Example Sheet 3 Q1 and Q2). ■

As an application of Lusin's theorem, let's prove density of $C_c(\mathbb{R}^d)$ in $L^1(\mathbb{R}^d)$:

Proof of Lemma 1.11. Fix $f \in L^1(\mathbb{R}^d)$ and $\varepsilon > 0$. For $r > 0$, set

$$E_r := \{x \in \mathbb{R}^d \colon x \in [-r, r]^d, |f(x)| \leq r\}$$

Note that $|f(x)\mathbb{1}_{E_r}(x)| \leq |f(x)|$ and $f(x)\mathbb{1}_{E_r}(x) \to f(x)$ for a.e. x. Thus, by the dominated convergence theorem, $||f\mathbb{1}_{E_r} - f||_1 \to 0$ as $r \to \infty$. Pick $R > 0$ sufficiently large such that $||f\mathbb{1}_{E_R} - f||_1 < \varepsilon/3$. Note that $|[-R, R]^d| < \infty$, so we can apply (weak) Lusin's theorem (Theorem 1.15) to the measurable function $f_R := (f\mathbb{1}_{E_R})|_{[-R,R]^d}$ defined on $[-R, R]^d$. We thus obtain a closed subset $K \subset [-R, R]^d$ such that $|[-R, R]^d \backslash K| < \varepsilon/6R$ and $f_R|_K$ is continuous. Note that K is also closed in $\mathbb{R}^d$. Thus, by Tietze's extension theorem, there exists $g \in C(\mathbb{R}^d)$ such that $g|_K = f_R|_K$ and $||g||_\infty = ||f_R|_K||_\infty = R$. Now, set $B := \mathbb{R}^d \backslash (-R - \delta, R + \delta)^d$, where $\delta > 0$ is chosen sufficiently small such that $|(-R - \delta, R + \delta)^d \backslash [-R, R]^d| < \varepsilon/3R$. Note that B and K are both closed, so by Urysohn's lemma, there exists $\phi \colon \mathbb{R}^d \to [0, 1]$ such that $\phi|_B = 0$ and $\phi|_K = 1$. Set $h = g\phi$. Note that $||h||_\infty \leq ||g||_\infty$ by construction. Finally, observe that $h \in C_c(\mathbb{R}^d)$ and

$$\begin{aligned}
||h - f||_1 &\leq ||f\mathbb{1}_{E_R} - f||_1 + \int_{[-R,R]^d \backslash K} |h(x) - f(x)\mathbb{1}_{E_R}(x)|\, dx \\
&\quad + \int_{(-R-\delta,R+\delta)^d \backslash [-R,R]^d} |h(x) - f(x)\mathbb{1}_{E_R}(x)|\, dx \\
&< \frac{\varepsilon}{3} + \left|[-R, R]^d \backslash K\right| \sup_{[-R,R]^d \backslash K} |h - f\mathbb{1}_{E_R}| \\
&\quad + \left|(-R - \delta, R + \delta)^d \backslash [-R, R]^d\right| \sup_{(-R-\delta,R+\delta)^d \backslash [-R,R]^d} |h| \\
&< \frac{\varepsilon}{3} + \frac{\varepsilon}{6R} \cdot 2R + \frac{\varepsilon}{3R} \cdot R = \varepsilon
\end{aligned}$$

as required. ■

Finally, the third of Littlewood's principles is made rigorous by Egorov's theorem.

Theorem 1.17 Egorov's theorem
Let $(E, \mathcal{B}, \mu)$ be a finite measure space, and let $(f_n)_{n\in\mathbb{N}}$ be a sequence of measurable functions such that $f_n(x) \to f(x)$ for μ-a.e. $x \in E$. Then for every $\varepsilon > 0$, there exists a set $A \in \mathcal{B}$ such that $(f_n|_A)_{n\in\mathbb{N}}$ converges uniformly and $\mu(E \backslash A) < \varepsilon$.

Proof. For each $k, N \in \mathbb{N}$, define

$$A_{k,N} = \left\{x \in E \colon |f(x) - f_n(x)| < \frac{1}{k}\ \forall\, n \geq N\right\}$$

Note that $(A_{k,N})_{N\in\mathbb{N}}$ is an increasing sequence, with

$$\bigcup_{N\in\mathbb{N}} A_{k,N} = \{x \in E \colon f_n(x) \to f(x)\}$$

Thus, we have $\mu\left(E \backslash \bigcup_{N\in\mathbb{N}} A_{k,N}\right) = 0$. Since $(E \backslash \bigcup_{N=1}^n A_{k,N})_{n\in\mathbb{N}}$ is a decreasing sequence with $\mu(E) < \infty$, we have $\mu\left(E \backslash \bigcup_{N=1}^n A_{k,N}\right) \downarrow 0$ and thus, for each $k \in \mathbb{N}$, we can pick $n(k)$ such that

$$\mu\left(E \backslash \bigcup_{N=1}^{n} A_{k,N}\right) < \frac{\varepsilon}{2^k}$$

Now, define $A = \bigcap_k A_{k,n(k)}$. Observe that $\mu(E \backslash A) < \sum_{k=1}^\infty \frac{\varepsilon}{2^k} = \varepsilon$ and that $|f_N(x) - f(x)| < 1/k$ for all $N \geq n(k)$ and $x \in A$, so $(f_n|_A)_{n\in\mathbb{N}}$ converges uniformly. ■

1.5 Riesz representation theorem

Let E be a compact metric space and $\mathcal{B}$ the Borel σ-algebra. We denote by $C(E)$ the space of continuous functions on E. Recall from II Linear Analysis that this is a Banach space when endowed with the norm $||\cdot||_\infty$.

A *linear functional* is a map $L\colon C(E) \to \mathbb{C}$ such that $L(a_1 f_1 + a_2 f_2) = a_1 L(f_1) + a_2 L(f_2)$ for all $a_1, a_2 \in \mathbb{C}$ and $f_1, f_2 \in C(E)$. We say that a linear functional L is *bounded* if there exists $C > 0$ such that $|L(f)| \leq C||f||_\infty$ for all $f \in C(E)$. We say that a linear functional L is *positive* if $L(f) \geq 0$ for all $f \in C(E)$ with $f(E) \subset \mathbb{R}_{\geq 0}$.

Example
Let E be a compact metric space. For any finite Borel measure μ,

$$L(f) = \int_E f \, d\mu$$

defines a bounded positive linear functional on $C(E)$.

Theorem 1.18 Riesz representation theorem
Let E be a compact metric space, and let L be a positive bounded linear functional on $C(E)$. Then there is a unique finite Borel measure μ such that

$$L(f) = \int f \, d\mu$$

for all $f \in C(E)$. Moreover, $||L|| = \mu(E)$.

There is a version of the theorem for bounded linear functionals that may not be positive.

Theorem 1.19 Riesz representation theorem
For every bounded linear function L on $C(E)$, there exists a unique complex Borel measure μ such that

$$L(f) = \int f \, d\mu$$

for all $f \in C(E)$. Moreover, $||L|| = ||\mu||$, where $||\mu||$ is the total variation norm of the μ.

Proof of uniqueness. Let μ_1, μ_2 be finite measures such that $\int f \, d\mu_1 = \int f \, d\mu_2$ for all $f \in C(E)$. Fix $A \in \mathcal{B}$. Pick K_1, K_2 compact and U_1, U_2 open such that $K_j \subset A \subset U_j$ and $\mu_j(U_j \backslash K_j) < \varepsilon$ for $j \in \{1, 2\}$. Set $U = U_1 \cap U_2$ and $K = K_1 \cup K_2$. Then $K \subset A \subset U$ and $\mu_j(U \backslash K) < \varepsilon$ for $j \in \{1, 2\}$. By Urysohn, we can pick $g \in C(E)$ such that $g|_K = 1$ and $g|_{E \backslash U} = 0$. Then

$$\left| \int \mathbb{1}_A \, d\mu_j - \int g \, d\mu_j \right| < \varepsilon$$

Since $\int g \, d\mu_1 = \int g \, d\mu_2$, it follows that

$$|\mu_1(A) - \mu_2(A)| < \varepsilon.$$

Taking $\varepsilon \to 0$, we deduce that $\mu_1(A) = \mu_2(A)$. Hence, $\mu_1 = \mu_2$. ■

We will not be proving existence in this course.

2 Lebesgue spaces

2.1 Definitions and basic properties

Let $(E, \mathcal{B}, \mu)$ be a measure space. For a measurable function $f\colon E \to \mathbb{C}$ and $p \in [1, \infty)$, we write

$$||f||_p := \left(\int_E |f|^p \, d\mu \right)^{1/p}$$

and

$$||f||_\infty := \inf\{t \geq 0 \colon |f(x)| \leq t \text{ for } \mu\text{-a.e. } x \in E\}$$

where, by convention, the infimum of an empty set is taken to be ∞. For $p \in [1, \infty]$, we denote by $\mathcal{L}^p(E, \mathcal{B}, \mu)$ the set of all measurable functions $E \to \mathbb{C}$ for which $|||f||_p < \infty$. Writing $f \sim g \iff f = g$ a.e., we then define the Lebesgue space $L^p(E, \mathcal{B}, \mu) := \mathcal{L}^p(E, \mathcal{B}, \mu)/\sim$. Moreover, in the case that E is a topological space, there is also a notion of *local* Lebesgue spaces:

Definition Local Lebesgue space L^p_{loc}
Let E be a topological space and μ a Borel measure. For $p \in [1, \infty]$, we define

$$L^p_{\text{loc}}(E) := \{f \colon E \to \mathbb{C} \colon f\mathbb{1}_K \in L^p(E) \ \forall \text{ compact } K \subset E\}$$

Recall the Minkowski inequality: if $f, g \in L^p$, then $f + g \in L^p$ with

$$||f + g||_p \leq ||f||_p + ||g||_p$$

It is then easy to check that L^p is a Banach space for $p \in [1, \infty]$.

Finally, we also recall Hölder's inequality: if $f \in L^p(E)$ and $g \in L^q(E)$ for $p, q \in [1, \infty]$ that satisfy $1/p + 1/q = 1$, then $fg \in L^1(E)$ with

$$||fg||_1 \leq ||f||_p ||g||_q$$

Now, let us begin with a generalisation of Minkowski's inequality:

Theorem 2.1 Minkowski's integral inequality
Let $(E, \mathcal{A}, \mu)$ and $(F, \mathcal{B}, \nu)$ be σ-finite measure spaces and $p \in [1, \infty]$. Suppose $G \colon E \times F \to \mathbb{R}$ are $\mathcal{A} \otimes \mathcal{B}$-measurable. If $\int_F ||G(\cdot, y)||_p \, d\nu(y) < \infty$, then

$$g(x) = \int G(x, y) \, d\nu(y)$$

exists for μ-a.e. x and

$$||g||_p \leq \int ||G(\cdot, y)||_p \, d\nu(y)$$

Remark. Note that, if $F = \{1, 2\}$ with counting measure ν, then the above reduces to the usual Minkowki inequality.

Proof of Theorem 2.1. The result is vacuously true when g vanishes μ-a.e., so we may assume WLOG that this is not the case.

For $p = \infty$, note that $|G(x, y)| \leq ||G(\cdot, y)||_\infty$ for $\mu \otimes \nu$-a.e. (x, y), so

$$|g(x)| \leq \int |G(x, y)| \, d\nu(y) \leq \int ||G(\cdot, y)||_\infty \, d\nu(y)$$

for μ-a.e. $x \in E$.

Now, suppose $p < \infty$. If $p = 1$, the result is immediate from Fubini's theorem, so it remains to carefully consider the case $p \in (1, \infty)$. Observe that

$$\begin{aligned}\int_E \left| \int_F G(x,y)\, d\nu(y) \right|^p d\mu(x) &\leq \int_E \left(\int_F |G(x,y)|\, d\nu(y) \right) |g(x)|^{p-1}\, d\mu(x) \\ &= \int_F ||G(\cdot, y) g^{p-1}||_1\, d\nu(y) \\ &\leq \int_F ||G(\cdot, y)||_p ||g^{p-1}||_{p/(p-1)}\, d\nu(y) \\ &= ||g||_p^{p-1} \int_F ||G(\cdot, y)||_p\, d\nu(y)\end{aligned}$$

Dividing both sides by $||g||_p^{p-1}$ yields the result. ■

For the rest of this subsection, we specialise to $E = \mathbf{R}^d$ equipped with the Lebesgue measure.

Proposition 2.2
For every $p \in [1, \infty)$, $C_c(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d)$.

Proof. The proof is identical to the $p = 1$ case. ■

Proposition 2.3
The space of $L^p(\mathbb{R}^d)$ is separable for all $p \in [1, \infty)$.

Proof. Let $p \in [1, \infty)$. Fix $f \in L^p(\mathbb{R}^d)$ and $\varepsilon > 0$. Let

$$\mathcal{R}_{\mathbb{Q}} := \left\{ \prod_{i=1}^d (a_i, b_i] \colon a_i, b_i \in \mathbb{Q} \right\}$$

and

$$S_{\mathbb{Q}} := \left\{ s = \sum_{k=1}^N (\alpha_k + i\beta_k) \mathbb{1}_{R_k} \colon N \in \mathbb{N}, \alpha_k, \beta_k \in \mathbb{Q}, R_k \in \mathcal{R}_k \right\}$$

Note that $S_{\mathbb{Q}}$ is countable. We would then be done if we can show that $S_{\mathbb{Q}}$ is dense in $L^p(\mathbb{R}^d)$.

Fix $f \in L^p(\mathbb{R}^d)$ and $\varepsilon > 0$. By density of $C_c(\mathbb{R}^d)$ in $L^p(\mathbb{R}^d)$, we can pick $h \in C_c(\mathbb{R}^d)$ such that $||h - f||_p < \varepsilon/2$. We can then pick $K > 0$ such that $\operatorname{supp} h \subset [-K/2, K/2]^d$. Choose $0 < \eta < \varepsilon/2K^d$. Since h is continuous, we can pick $\delta > 0$ such that $|x - y| < \delta \implies |h(x) - h(y)| < \eta$. Pick $N \in \mathbb{N}$ sufficiently large such that $N^d \geq 1/\delta$. Partition $[-K/2, K/2]$ into a finite union of cubes $\{C_k\}_{k=1}^N$ from $\mathcal{R}_{\mathbb{Q}}$ of side length less than $1/N$. For each cube C_k in the partition, pick $\alpha_k, \beta_k \in \mathbb{Q}$ such that $|\alpha_k + i\beta_k - h(x)| < \eta$ for all $x \in C_k$. Now, observe that

$$\left|\left| \sum_{k=1}^N (\alpha_k + i\beta_k) \mathbb{1}_{C_k} - h \right|\right|_p \leq \sum_{k=1}^N ||(\alpha_k + i\beta_k - h)\mathbb{1}_{C_k}||_p < K^d \eta < \frac{\varepsilon}{2}$$

Hence,

$$\left|\left| \sum_{k=1}^N (\alpha_k + i\beta_k) \mathbb{1}_{C_k} - f \right|\right|_p < \varepsilon$$

as required. ■

For $a \in \mathbb{R}^d$, denote by τ_a the translation-by-a operator, $\tau_a f(x) = f(x - a)$.

Proposition 2.4
Let $p \in [1, \infty)$ and $f \in L^p(\mathbb{R}^d)$. Then the map

$$\begin{aligned} \mathbb{R} &\to L^p(\mathbb{R}^d) \\ a &\mapsto \tau_a f \end{aligned}$$

is continuous.

Proof. Fix $\varepsilon > 0$. Pick $h \in C_c(\mathbb{R}^d)$ such that $||h - f||_1 < \varepsilon/3$. Note that h is uniformly continuous, so for every $\eta > 0$, there exists $\delta > 0$ such that $|x - y| < \delta \implies |h(x) - h(y)| < \eta$. Then, for any $a, b \in \mathbb{R}^d$ with $|a - b| < \delta$, we have

$$|\tau_a h(x) - \tau_b h(x)| = |h(x - a) - h(x - b)| < \eta$$

Note that $|\tau_a h(x) - \tau_b h(x)| > 0$ only if $x \in \operatorname{supp} \tau_a h \cup \operatorname{supp} \tau_b h$. The measure of this set is bounded above by $C = C(h)$. Thus, we can estimate

$$||\tau_a h - \tau_b h||_p \leq \eta C^{1/p}$$

from which it follows that

$$\begin{aligned} ||\tau_a f - \tau_b f||_p &\leq ||\tau_a f - \tau_a h||_p + ||\tau_a h - \tau_b h||_p + ||\tau_b h - \tau_b f||_p \\ &= 2||f - h||_p + ||\tau_a h - \tau_b h||_p \\ &< \frac{2\varepsilon}{3} + \eta C^{1/p} \end{aligned}$$

Picking η sufficiently small, we then get $||\tau_a f - \tau_b f||_p < \varepsilon$, as required. ■

Theorem 2.5
Let $(E, \mathcal{B}, \mu)$ be a σ-finite measure space. Let $p \in [1, \infty)$ and let q be defined by $p^{-1} + q^{-1} = 1$. Then for all $g \in L^q(E)$, the map

$$L_g(f) = \int fg \, d\mu$$

is a bounded linear functional on $L^p(E)$ with $||L_g|| = ||g||_q$. Conversely, for every bounded linear functional L on $L^p(E)$, there is a unique $g \in L^q(E)$ such that $L = L_g$.

Remark. Taking $E = \mathbb{N}$ equipped with counting measure, we recover the familiar result $\ell_q' \equiv \ell_p$ from II Linear Analysis.

Proof. The fact that $L_g \in L^p(E)'$ with $||L_g|| \leq ||g||_q$ is immediate from linearity of integration and Hölder's inequality. Now, let us show that $||L_g|| = ||g||_q$. If $p > 1$, set

$$f(x) = \begin{cases} g(x)^{-1}|g(x)|^q, & g(x) \neq 0 \\ 0, & g(x) = 0 \end{cases}$$

and observe that $||f||_p^p = ||g^{p(q-1)}||_1 = ||g||_q^q$ and $L_g(f) = ||g||_q^q = ||f||_p ||g||_q$. In the case $p = 1, q = \infty$, we want to show that the set $A := \{x \in E \colon |g(x)| > ||L||\}$ has measure zero. Suppose, on the contrary, that $\mu(A) > 0$. Set

$$f(x) = \begin{cases} g(x)^{-1}, & x \in A \\ 0, & x \notin A \end{cases}$$

and note that $||f||_1 < ||L||^{-1}\mu(A)$ and $L(f) = \int fg \, d\mu = \mu(A)$. Thus, $\mu(A) \leq ||L_g|| ||f||_1 < \mu(A)$ — a contradiction!

Now, we prove the converse. Uniqueness follows easily from the fact that $||L_{g_1} - L_{g_2}|| = ||L_{g_1 - g_2}|| = ||g_1 - g_2||_q$. For existence, fix a bounded linear function L on $L^p(E)$. We will first prove the result for finite measure space and extend it to the σ-finite case.

CASE 1: $\mu(E) < \infty$

In this case, $L^\infty(E) \subset L^p(E)$, and we may define the function $\nu(A) = L(\mathbb{1}_A)$ for $A \in \mathcal{B}$. This is finitely additive because L is linear. To show that ν is σ-additivity, it suffices to show that $\lim_{n\to\infty} \nu(A_n) = 0$ for every decreasing sequence $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{B}$ with $\bigcap_n A_n = \varnothing$. Indeed,

$$|\nu(A_n)| = |L(\mathbb{1}_A)| \leq ||L|| ||\mathbb{1}_A||_p = ||L||\, \mu(A)^{1/p} \to 0$$

Thus, we have now shown that ν is a complex measure.

Now, note that $\nu(A) = 0 \implies \mathbb{1}_A = 0$ a.e. $\implies L(1_{\mathbb{A}}) = 0 \implies \nu(A) = 0$, so $\nu \ll \mu$. By the Radon-Nikodym theorem (Theorem 1.6), we can pick $g \in L^1(E)$ such that

$$L(\mathbb{1}_A) = \nu(A) = \int \mathbb{1}_A g \, d\mu$$

By linearity of L and integration, we have that $L(f) = \int fg\, d\mu$ for all simple functions f. By density of simple functions in $L^\infty(E)$, the same holds for all $f \in L^\infty(E)$. Finally, density of $L^\infty(E)$ in $L^p(E)$ gives us $L(f) = \int fg\, d\mu$ for all $f \in L^p(E)$.

To complete the proof in this case, it suffices to show that $g \in L^q(E)$. If $p > 1$, consider

$$g_R(x) = \begin{cases} g(x), & |g(x)| \leq R \\ R\,\mathrm{sign}(g(x)), & \text{otherwise} \end{cases}$$

and let

$$f_R(x) = \begin{cases} g_R(x)^{-1} g_R(q)^q, & g_R(x) \neq 0 \\ 0, & g_R(x) = 0 \end{cases}$$

Note that $||f_R||_p = ||g_R||_q^{q-1}$ and that

$$L(f_R) = \int f_R g\, d\mu \geq \int f_R g_R\, d\mu \geq ||g_R||_q^q$$

Thus, $||g_R||_q^q \leq ||g_R||_q^{q-1} ||L||$ and so $||g_R||_q \leq ||L||$. Taking $R \to \infty$, we get that $g \in L^q$ with $||g||_q \leq ||L||$.

CASE 2: μ is σ-finite

In this case, we can pick an increasing sequence $(E_n)_{n\in\mathbb{N}}$ such that $\bigcup_n E_n = E$ and $\mu(E_n) < \infty$ for each $n \in \mathbb{N}$. Note that there are natural inclusions $L^p(E_1) \subset L^p(E_2) \subset \cdots \subset L^p(E)$ by identifying a function on a smaller subset with a function on the larger subset but which vanishes outside the smaller subset. Applying the result of Case 1, we can pick $g_n \in L^p(E_n)$, for each n, such that $L|_{L^p(E_n)} = L_{g_n}$. By injectivity of $g \mapsto L_g$, we have that $g_m|_{E_n} = g_n$ whenever $m > n$. It follows that we can define $g\colon E \to \mathbb{C}$ by setting $g(x) = g_n(x)$ whenever $x \in E_n$. It is measurable since each g_n is measurable and $E = \bigcup_n E_n$. Since $||g_n||_q \leq ||L||$, we also have that $||g||_q \leq ||L||$ by the monotone convergence theorem. By construction, $L(f) = L_g(f)$ for all $f \in \bigcup_n L^p(E_n)$. Note that this space is dense in $L^p(E)$ (since each $f \in L^p(E)$ can be approximated by $f\mathbb{1}_{E_n} \in L^p(E_n) \subset L^p(E)$). Hence, we in fact have $L(f) = L_g(f)$ for all $f \in L^p(E)$. ■

Remark. Note the technique used in the proof to prove σ-additivity, namely that it suffices to show finite additivity and that $\lim_{n\to\infty} \nu(A_n) = 0$ for every decreasing sequence $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{B}$ with $\bigcap_n A_n = \varnothing$. Indeed, let $(E_n)_{n\in\mathbb{N}}$ be a sequence of pairwise disjoint sets in $\mathcal{B}$. Set $S_N = \bigcup_{n=1}^N E_n$. By finite additivity, $\mu(S_N) = \sum_{n=1}^N \mu(E_n)$. Note also that $(S_\infty \backslash S_N)_{N\in\mathbb{N}}$ is a decreasing sequence with $\bigcap_N (S_\infty \backslash S_N) = \varnothing$. Thus, by the second condition, we have $\mu(S_\infty) = \lim_{N\to\infty} \mu(S_N) = \sum_{n=1}^\infty \mu(E_n)$, as required.

2.2 Convolutions and mollifications

Definition Convolution
Let f, g be measurable functions $\mathbb{R}^d \to \mathbb{C}$. We define their convolution to be

$$(f \star g)(x) = \int f(t)g(x-t)\,dt$$

if the above integral exists for a.e. $x \in \mathbb{R}^d$.

Lemma 2.6
Let f, g be measurable functions $\mathbb{R}^d \to \mathbb{C}$.
(i) *$(f \star g)(x) = (g \star f)(x)$ if the integral exists at x*
(ii) *For any $a \in \mathbb{R}^d$, $\tau_a(f \star g) = \tau_a(f) \star g = f \star \tau_a(g)$ provided the convolutions exist.*

Now, when do convolutions exist and, if they do, how regular are they?

Theorem 2.7 Young's inequality
Let $p, q, r \in [1, \infty]$ satisfy $p^{-1} + q^{-1} = r^{-1} + 1$. If $f \in L^p(\mathbb{R}^d)$ and $g \in L^q(\mathbb{R}^d)$, then the integral $f \star g$ exists for a.e. x. Moreover, $f \star g \in L^r(\mathbb{R}^n)$ with the estimate

$$||f \star g||_r \leq ||f||_p ||g||_q$$

In lectures, we will only prove two special cases.

Proof for $q = 1, p = r$. Consider the map

$$\begin{aligned} G\colon \mathbb{R}^d \times \mathbb{R}^d &\to \mathbb{C} \\ (x,t) &\mapsto g(t)f(x-t) \end{aligned}$$

Note that this is measurable and that

$$\int ||G(\cdot,t)||_p\,dt = \int |g(t)|||\tau_t f||_p\,dt = ||f||_p||g||_1 < \infty$$

Thus, by Minkowski's integral inequality (Theorem 2.1), we have that $g \star f$ exists for a.e. $x \in X$ (and thus $f \star g$ as well). Moreover, we have the estimate

$$||f \star g||_r \leq \int ||G(\cdot,t)||_p\,dt = ||f||_p||q||_1$$

as required. ■

Proof for $r = \infty$. Since $p^{-1} + q^{-1} = 1$, we have

$$|(f \star g)(x)| \leq \int |f(t)g(x-t)|\,dt \leq ||f||_p||q||_q$$

by Hölder's inequality. ■

Remark. Note that, in the $r = \infty$ case, the proof actually gives us a stronger result! In this case, the integral exists and the inequality holds for all $x \in \mathbb{R}^d$, not just a.e.

Lemma 2.8
Let $f, g \in L^1(\mathbb{R}^d)$ and $h \in L^\infty(\mathbb{R}^d)$. Then

$$(f \star g) \star h = f \star (g \star h)$$

Proof. Existence of the convolutions follows immediately from Young's inequality (Theorem 2.7). Observe that

$$[(f \star g) \star h](x) = \int (f \star g)(t) h(x-t)\, dt = \int \left[\int f(s) g(t-s)\, ds \right] h(x-t)\, dt$$

Now, note that

$$\int \int |f(s) g(t-s) h(x-t)|\, dt\, ds \leq ||f||_1 ||g||_1 ||h||_\infty < \infty$$

We can thus use Fubini to deduce that

$$\begin{aligned}
[(f \star g) \star h](x) &= \iint f(s) g(t-s) h(x-t)\, dt\, ds \\
&= \iint f(s) g(u) h(x-s-u)\, du\, ds \\
&= \int f(s) (g \star h)(x-s)\, ds \\
&= [f \star (g \star h)](x)
\end{aligned}$$

as required. ■

> **Remark.** The result still holds under the conditions $f \in L^p(\mathbb{R}^d), g \in L^q(\mathbb{R}^d), h \in L^r(\mathbb{R}^d)$ if $p^{-1} + q^{-1} + r^{-1} \geq 2$.

Lemma 2.9
Let $p, q \in [1, \infty]$ satisfy $p^{-1} + q^{-1} = 1$. If $f \in L^p(\mathbb{R}^d)$ and $g \in L^q(\mathbb{R}^d)$, then $f \star g \in C(\mathbb{R}^d)$.

Proof. By symmetry in p and q, we may assume WLOG that $q < \infty$. By Young's inequality (Theorem 2.7) and the remark after the proof for $r = \infty$, we see that, for every $x, y \in \mathbb{R}^d$,

$$\begin{aligned}
|(f \star g)(x) - (f \star g)(y)| &= |(f \star g)(x) - [\tau_{y-x}(f \star g)](x)| \\
&= |[f \star (g - \tau_{y-x} g)](x)| \\
&\leq ||f||_p ||g - \tau_{y-x} g||_q
\end{aligned}$$

By continuity of translations in L^q (Proposition 2.4), we deduce that $|(f \star g)(x) - (f \star g)(y)| \to 0$ as $y \to x$, as required. ■

Notation *Multi-index notation*
Given $\alpha \in \mathbb{Z}^d_{\geq 0}$ and a function $f \colon \mathbb{R}^d \to \mathbb{C}$, we write

$$|\alpha| = \sum_{i=1}^d |\alpha_i|$$

and

$$D^\alpha = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}$$

whenever the partial derivative exists.

Definition Schwartz space
The Schwartz space on $\mathbb{R}^d$ is defined by

$$\mathcal{S}(\mathbb{R}^d) := \left\{ f \in C^\infty(\mathbb{R}^d) \colon \||x|^n D^\alpha f\|_\infty < \infty \right\}$$

for all $n \in \mathbb{Z}_{\geq 0}$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$.

Proposition 2.10
Let $f \in L^p(\mathbb{R}^d)$ for some $p \in [1, \infty]$. If $g \in \mathcal{S}(\mathbb{R}^d)$, then $f \star g \in C^\infty(\mathbb{R}^d)$ and

$$D^\alpha(f \star g) = f \star (D^\alpha g)$$

for every multi-index $\alpha \in \mathbb{Z}^d_{\geq 0}$.

Proof. By iteration, it suffices to show the result for $|\alpha| = 1$. Fix a multi-index α with $|\alpha| = 1$. Then we have

$$\begin{aligned}\frac{(f \star g)(x + t\alpha) - (f \star g)(x)}{t} - (f \star D^\alpha g)(x) &= \frac{[\tau_{-t\alpha}(f \star g)](x) - (f \star g)(x)}{t} - (f \star D^\alpha g)(x) \\ &= f \star \left(\frac{\tau_{-t\alpha} g - g}{t} - D^\alpha g \right)(x)\end{aligned}$$

For each $x \in \mathbb{R}^d$, the mean value theorem gives us $\tilde{s}(x)$ between 0 and t for which

$$\frac{(\tau_{-t\alpha})g(x) - g(x)}{t} = D^\alpha g(x + \tilde{s}(x)\alpha)$$

Apply the mean value theorem once more, we obtain $s(x)$ between 0 and $\tilde{s}(x)$ such that

$$\frac{(\tau_{-t\alpha})g(x) - g(x)}{t} - D^\alpha g(x) = \tilde{s}(x) D^{2\alpha} g(x + s(x)\alpha)$$

and thus

$$\left| \frac{(\tau_{-t\alpha})g(x) - g(x)}{t} - D^\alpha g(x) \right| \leq |t| \left| D^{2\alpha} g(x + s(x)\alpha) \right|$$

Next, we note that $|D^{2\alpha} g(x)| \leq C(|x| + 10)^{-d-1}$ holds for $C = \|(|x| + 10)^{d+1} D^{2\alpha} g(x)\|_\infty$ (which is finite since $g \in \mathcal{S}(\mathbb{R}^d)$). It then follows that

$$\left\| \frac{(\tau_{-t\alpha})g(x) - g(x)}{t} - D^\alpha g(x) \right\|_q \leq C|t| \, \|(|x + s(x)\alpha| + 10)^{-d-1}\|_q$$

For every $t \in (-1, 1)$, we then obtain the estimate

$$\left\| \frac{(\tau_{-t\alpha})g(x) - g(x)}{t} - D^\alpha g(x) \right\|_q \leq C|t| \, \|(|x| + 9)^{-d-1}\|_q$$

Noting that $\|(|x| + 9)^{-d-1}\|_q^q \leq \int_{\mathbb{R}^d} |x|^{-2q} \, dx \lesssim \int_0^\infty r^{-2q} \, dr < \infty$, we deduce that

$$\left\| \frac{(\tau_{-t\alpha})g(x) - g(x)}{t} - D^\alpha g(x) \right\|_q \to 0$$

and the result follows from Young's inequality (Theorem 2.7). ■

Definition Approximation of the identity
We say that a sequence of functions $(f_n)_{n \in \mathbb{N}}$ in $L^1(\mathbb{R}^d)$ is an approximation of the identity if

(i) $f_n(x) \geq 0$ *for all* $x \in \mathbb{R}^d$ *and all* $n \in \mathbb{N}$

(ii) $\displaystyle\int f_n \, dx = 1$ *for all* $n \in \mathbb{N}$

(iii) $\displaystyle\int_{|x| > \varepsilon} f_n(x) \, dx \to 0$ *for all* $\varepsilon > 0$

Lemma 2.11
Let $f \in L^1(\mathbb{R}^d)$ be nonnegative with $\int f\,dx = 1$. The sequence (f_n) defined by

$$f_n(x) = n^d f(nx)$$

is an approximation of the identity.

Proof. It remains to show (iii). By change of variables and dominated convergence, we have

$$\int_{|x|>\varepsilon} f_n(x)\,dx = \int_{|x|>n\varepsilon} f(x)\,dx \to 0$$

as required. ■

Theorem 2.12
Let (f_n) be an approximation of the identity. For every $g \in L^p(\mathbb{R}^d)$, we have $||f_n \star g - g||_p \to 0$ as $n \to \infty$.

Proof. WLOG, we may assume that $f_n(x) = 0$ on $|x| > 1$ for all $n \in \mathbb{N}$. To see why, consider

$$\tilde{f}_n(x) = \begin{cases} f_n(x)\left(\int_{|x|\le 1} f_n\,dx\right)^{-1}, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$$

It is easy to check that $||f_n - \tilde{f}_n||_1 \to 0$. By Young's inequality (Theorem 2.7), we then have $||f_n \star g - \tilde{f}_n \star g||_p \le ||f_n - \tilde{f}_n||_1 ||g||_p \to 0$. Thus, $||f_n \star g - g||_p \to 0$ iff $||\tilde{f}_n \star g - g||_p \to 0$.

Fix $\varepsilon > 0$. Pick $h \in C_c(\mathbb{R}^d)$ such that $||g - h|| < \varepsilon$. Since h is uniformly continuous, we can pick $\delta > 0$ such that $|x - y| < \varepsilon \implies |h(x) - h(y)| < \varepsilon$. Pick $N \in \mathbb{N}$ sufficiently large such that $\int_{|x|>\delta} f_N(x)\,dx < \varepsilon$. We then have

$$\begin{aligned} |(f_n \star h)(x) - h(x)| &\le \int f_n(t)|h(x-t) - h(x)|\,dt \\ &= \int_{|t|<\delta} f_n(t)|h(x-t) - h(x)|\,dt + \int_{|t|\ge\delta} f_n(t)|h(x-t) - h(x)|\,dt \\ &\le \varepsilon \int f_n(t)\,dt + 2||h||_\infty \int f_n(t)\,dt \\ &\le \varepsilon(1 + 2||h||_\infty) \end{aligned}$$

Pick $R > 0$ such that $\operatorname{supp} h \subset [-R, R]^d$. Then we have

$$\operatorname{supp}(h \star f_n) \subset \operatorname{supp} h + \operatorname{supp} f_n \subset [-R-1, R+1]^d$$

We then have that

$$||f_n \star h - h||_p \le [2(R+1)]^{d/p}(1 + 2||h||_\infty)\varepsilon$$

Hence, by the triangle inequality, we conclude that

$$\begin{aligned} ||f_n \star g - g||_p &\le ||f_n \star (g-h)||_p + ||f_n \star h - h||_p + ||h - g||_p \\ &\le 2||g-h||_p + |f_n \star h - h||_p \\ &= \left\{2 + [2(R+1)]^{d/p}(1 + 2||h||_\infty)\right\}\varepsilon \end{aligned}$$

as required. ■

Approximations of the identity can be particularly useful when they are C_c^∞. A standard

example would be to consider the function

$$\eta(x) = \begin{cases} C \exp\left(-\frac{1}{1-|x|^2}\right), & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$$

where C is chosen so that $\int \eta = 1$. By Lemma 2.11, we then get an approximation of the identity (η_n). This is sometimes called the *standard mollifier*.

Corollary 2.13
For every $p \in [1, \infty)$, $C_c^\infty(\mathbb{R}^d)$ is dense in $L^p(\mathbb{R}^d)$.

Proof. Fix $f \in L^p$ and $\varepsilon > 0$. Pick $R > 0$ sufficiently large such that $||f - f\mathbb{1}_{[-R,R]^d}|| < \varepsilon/2$. Set $g = f\mathbb{1}_{[-R,R]^d}$. By Theorem 2.12, we can pick $n \in \mathbb{N}$ such that $||g - g \star \eta_n|| < \varepsilon/2$. By Proposition 2.10, $g \star \eta_n \in C^\infty$. Since η_n is compactly supported, we have $g \star \eta_n \in C_c^\infty$. Finally, note that $||f - g \star \eta_n|| < \varepsilon$. ■

3 Hilbert and Banach space analysis

Recall from II Linear Analysis that a *normed space* is a vector space X over $\mathbb{C}$ equipped with a map $||\cdot||\colon X \to \mathbb{R}_{\geq 0}$ such that

(i) $||x|| \geq 0$ for all $x \in X$, with $||x|| = 0 \iff x = 0$

(ii) $||\lambda x|| = |\lambda|||x||$ for all $x \in X, \lambda \in \mathbb{C}$

(iii) $||x + y|| \leq ||x|| + ||y||$ for all $x, y \in X$

In this course, the field will always be $\mathbb{C}$ unless otherwise stated.

A normed space X that is complete is called a *Banach space*. If X is a Banach space equipped with a sesquilinear form $\langle \cdot, \cdot \rangle\colon X \times X \to \mathbb{C}$ such that $||x||^2 = \langle x, x \rangle$, then we say it is a *Hilbert space*.

A *linear functional* on a normed space X is a map $f\colon X \to \mathbb{C}$ such that $f(\lambda x + \mu y) = \lambda f(x) + \mu f(y)$ for all $\lambda \in \mathbb{C}$ and $x, y \in X$. We say that a linear functional $f\colon X \to \mathbb{C}$ is *bounded* if

$$||f|| := \sup_{x \in B_X} |f(x)| < \infty$$

We denote by X' the space of bounded linear functions $X \to \mathbb{C}$. It is a Banach space with respect to $||\cdot||$ as defined above.

Notation. We write $f(x) = \langle x, f \rangle$ for $f \in X'$ and $x \in X$.

A set-theoretic interlude

Let A be a non-empty set equipped with a relation $\preceq$. We say that $\preceq$ is a *partial order* if it is

(i) reflexive: $x \preceq x$ for all $x \in A$

(ii) antisymmetric: $x \preceq y$ and $y \preceq x$ implies $x = y$

(iii) transitive: $x \preceq y$ and $y \preceq z$ implies $x \preceq z$

In this case, we say that $(A, \preceq)$ is a partially ordered set. A subset $B \subset A$ is said to be *totally ordered* if at least one of $x \preceq y$ or $y \preceq x$ holds for all $x, y \in B$. We say that an element $m \in B$ is *maximal* in B if there does not exist $x \in B \backslash \{m\}$ such that $m \preceq x$. An element $x \in A$ is an *upper bound* for B if $y \preceq x$ holds for all $y \in B$.

Lemma 3.1 Zorn's lemma
Let $(A, \preceq)$ be a partially ordered set. If all totally ordered subsets $B \subset A$ has an upper bound in A, then A has at least one maximal element.

Proof. See II Logic and Set Theory. ■

3.1 Hahn–Banach theorem and its consequences

Theorem 3.2 Hahn-Banach theorem I
Let X be a normed space and $Y \subset X$ a subspace of X. For every $g \in Y'$, there exists $f \in X'$ such that $f|_Y = g$ and $||f|| = ||g||$.

An important consequence of the Hahn-Banach theorem is the existence of a norming functional.

Corollary 3.3
Let X be a normed space and $x_0 \in X \backslash \{0\}$. Then there exists $f \in S_{X^*}$ such that $f(x_0) = ||x_0||$.

Proof. Set $Y = \text{span}\{x_0\}$ and define $g(\lambda x_0) = \lambda ||x_0||$. Note that $g \in S_{Y^*}$ with $g(x_0) = ||x_0||$. Apply Hahn-Banach. ■

Remark. We call f in the corollary a *norming functional at* x_0. It shows that

$$||x|| = \max_{g \in B_{X^*}} |\langle x, g\rangle|$$

We will prove the following more general version of the Hahn-Banach theorem:

Theorem 3.4 Hahn–Banach theorem II
Let X be a $\mathbb{R}$-vector space. Suppose $p\colon X \to \mathbb{R}_{\geq 0}$ satisfies
(i) *positive homogeneity: $p(\lambda x) = \lambda p(x)$ for all $\lambda > 0$ and $x \in X$*
(ii) *subadditive: $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$*
Let Y be subspace of X, and let $g \in Y'$ such that $g(x) \leq p(x)$ for all $x \in X$. Then there exists $f \in X'$ such that $f|_Y = g$ and $f(x) \leq p(x)$ for all $x \in X$.

Remark. An explanation is in order regarding the fact that the second version involves $\mathbb{R}$ as the field. Note that $\operatorname{Re} g\colon Y \to \mathbb{R}$ is a real linear functional when regarding Y as a real vector space. Note that, writing $p(x) := ||x||||g||$, we have $(\operatorname{Re} g)(x) \leq p(x)$, so we can use Hahn–Banach II to obtain a real linear functional $\tilde{f}\colon X \to \mathbb{R}$ satisfying $\tilde{f}(x) \leq p(x)$. Let $f(x) = \tilde{f}(x) - i\tilde{f}(ix)$. One can check that f is a complex linear functional such that $f|_Y = g$ and $||f|| = ||g||$.

Proof of Theorem 3.4. We will proceed in two steps: first extend in one direction and proceed via Zorn's lemma.

Step 1: Extension in one direction

Fix $x \in X \backslash Y$. For each $a \in \mathbb{R}$, define $g_a(y + \lambda x) = g(y) + a\lambda$ for all $\lambda \in \mathbb{R}$ and $y \in Y$. This is a linear functional on Y^+ that extends g. Fix $z \in Y$. Note that, for every $y \in Y$, we have

$$\begin{aligned} g(y) + g(z) = g(y + z) \leq p(y + z) \leq p(y - x) + p(z + x) \\ \implies g(y) - p(y - x) \leq p(z + x) - g(z) \end{aligned}$$

Set $a := \sup_{y \in Y} g(y) - p(y - x)$, which exists by the above inequality. It remains to show that $g_a \leq p$ on X. For $\lambda > 0$, we can take $z = \lambda^{-1} y$ and we have

$$a \leq p(\lambda^{-1} y + x) - g(\lambda^{-1} y) \implies g_a(y + \lambda x) = g(y) + a\lambda \leq p(y + \lambda x)$$

If $\lambda < 0$, then observe that

$$g(-\lambda^{-1} y) - p(-\lambda^{-1} y - x) \leq a \implies g_a(y + \lambda x) = g_a(y) + \lambda a \leq p(y + \lambda x)$$

which completes the first step.

Step 2: Extension to all of X

Let U be the collection of all partial extensions of g to a subspace of X that remain bounded by p; that is, each element $h \in U$ is a linear functional $h\colon X_h \to \mathbb{R}$ for some subspace $X_h \subset X$ such that $h|_Y = g$ and $h(x) \leq p(x)$ for all $x \in X_h$. Define the relation $\preceq$ on U by

$$h_1 \preceq h_2 \iff h_2 \text{ extends } h_1$$

It is easy to check that this is a partial order on U. We would like to show that Zorn's lemma applies here. Fix a totally ordered subset $V \subset U$. Let $X_V = \bigcup_{h \in V} X_h$, which one can verify is a subspace of X. Define $h_V\colon X_V \to \mathbb{R}$ by $x \mapsto h(x)$ for some

(equivalently, any) $h \in V$ such that $x \in X_h$. It is easy to check that this is well-defined, extends g to X_V and satisfies $h_V \leq p$. Moreover, observe that h_V is an upper bound for V. ■

The Hahn-Banach theorem has important geometric consequences, namely the separation theorems.

Theorem 3.5
Let X be a real normed space. Let $A, B \subset X$ be disjoint convex subsets with A open. Then there exists $f \in X'$ and $\lambda \in \mathbb{R}$ such that $f(A) \subset (-\infty, \lambda)$ and $f(B) \subset [\lambda, \infty]$.

Remark. Geometrically, this says that the hyperplane $\{x \colon f(x) = \lambda\}$ separates A and B.

Corollary 3.6 Mazur's theorem
Let X be a real normed space. If $Y \subset X$ is a subspace and $A \subset X$ is an open convex subset disjoint from Y, then there exists $f \in X'$ such that $f(Y) = \{0\}$ and $f(A) \subset (-\infty, 0)$.

Remark. Geometrically, this says that there is a hyperplane containing Y that is disjoint from A.

Proof. Apply Theorem 3.5 with $B = Y$ to obtain $f \in X'$ and $\lambda \in \mathbb{R}$ such that $f(A) \subset (-\infty, \lambda)$ and $f(Y) \subset [\lambda, \infty]$. Since Y is a subspace of X, $f(Y)$ is a subspace of $\mathbb{R}$. We cannot have $f(Y) = \mathbb{R}$, so $f(Y) = \{0\}$ and thus $\lambda < 0$. Hence, we also have $f(A) \subset (-\infty, \lambda) \subset (-\infty, 0)$. ■

Corollary 3.7
Let Y be a real normed space. If $Y \subset X$ be a subspace that is not dense and $x \in X \backslash \overline{Y}$, then there exists $f \in X'$ such that $f(Y) = \{0\}$ and $f(x) \neq 0$.

Proof. Let A be an open ball around x that is disjoint from Y. Apply the previous corollary. ■

Proof of Theorem 3.5. We will first prove the special case that B is a single point and subsequently extend it to a general convex subset.

STEP 1: $B = \{y\}$ is a single point

By translation invariance, we may assume WLOG that $0 \in A$. Define the *Minkowski functional*

$$p(x) = \inf\{\lambda > 0 \colon p(\lambda^{-1}x) \in A\}$$

You will show on Example Sheet 2 that it satisfies

- $p(\lambda x) = \lambda p(x)$ for all $\lambda > 0$ and $x \in X$
- $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$

Next, consider the map

$$\begin{aligned} g \colon \operatorname{span}_{\mathbb{R}}\{y\} &\to \mathbb{R} \\ \lambda y &\mapsto \lambda \end{aligned}$$

Observe that $y \notin A \implies p(y) \geq 1 = g(y)$. By homogeneity, we have that $p(\lambda y) \geq g(\lambda y)$ for all $\lambda > 0$ and, of course, we also have that $p(\lambda y) > 0 \geq g(\lambda y)$ for $\lambda \geq 0$. We can then apply the Hahn-Banach theorem to obtain $f \in X'$ with $f(x) \leq p(x)$ for all $x \in X$ that extends g. For all $x \in A$, we have that $f(x) \leq p(x) < 1$. On the other hand, $f(y) = g(y) = 1$.

STEP 2: Note that $\tilde{A} := A - B$ is simply the union of translates of A by elements of $-B$, so it is open. It is also easy to verify that $\tilde{A}$ is still convex. Applying Step 1 to $\tilde{A} = A - B$ and $\tilde{B} = \{0\}$ yields $f \in X'$ and $\lambda \in \mathbb{R}$ such that $f(\tilde{A}) \subset (-\infty, \lambda)$ and $0 = f(0) \geq \lambda$. Therefore, $f(\tilde{A}) \subset (-\infty, 0)$ and $f(x - y) < 0$ for all $x \in A, y \in B$. Set $\lambda = \sup\{f(x) \colon x \in A\}$ and note that $f(y) \geq \lambda$ for all $y \in B$. By a question on Example Sheet 2, you will see that $f(A)$ is open since A is open. Thus, it follows that $f(A) \subset (-\infty, \lambda]$ implies that $f(A) \subset (-\infty, \lambda)$. ■

3.2 Strong, weak and weak-$*$ topologies

Recall from II Linear Analysis that the closed unit ball B_X in a normed space X is compact iff $\dim X < \infty$. However, it is often useful to have a notion of compactness in order to, for instance, obtain a minimiser in the calculus of variations. To restore a similar notion, we consider other topologies on X.

Definition Seminorm
Let X be a $\mathbb{C}$-vector space. A seminorm is a map $p \colon X \to [0, \infty)$ such that
(i) $p(ax) = |a|p(x)$ *for all* $a \in \mathbb{C}$ *and* $x \in X$
(ii) $p(x + y) \leq p(x) + p(y)$ *for all* $x, y \in X$

Remark. This is clearly weaker than the notion of a norm as we do not require that $p(x) > 0$ for $x \neq 0$.

Given a family of seminorms $\mathcal{P}$, we can define a topology on X as follows. For $n \in \mathbb{N}$, $p_1, \cdots, p_n \in \mathcal{P}$ and $\varepsilon > 0$, consider the neigbourhood of some $x \in X$ given by

$$N(x; p_1, \cdots, p_n, \varepsilon) := \{y \in X \colon p_j(x - y) < \varepsilon \ \forall j \in \{1, \cdots, n\}\}$$

This defines a local base at x for some topology called the *topology induces by* $\mathcal{P}$. A set $U \subset X$ is open in this topology iff, for every $x \in U$, there exist $n \in \mathbb{N}, p_1, \cdots, p_n \in \mathcal{P}$ and $\varepsilon > 0$ such that $N(x; p_1, \cdots, p_n, \varepsilon) \subset U$.

Exercise. Check that the collection defined above is indeed a topology on X.

Definition Separating family
We say that a family $\mathcal{P}$ of seminorms is separating if, for every $x \in X$, there exists $p \in \mathcal{P}$ such that $p(x) > 0$.

Lemma 3.8
If $\mathcal{P}$ is a separating family of seminorms on a vector space X, then the topology induced by $\mathcal{P}$ is Hausdorff.

Proof. Fix distinct elements $x, y \in X$. Then there exists $p \in \mathcal{P}$ such that $p(x - y) > 0$. Set $\varepsilon = p(x - y)/2$. Then $N(x; p, \varepsilon)$ and $N(y; p, \varepsilon)$ are disjoint open subsets. ■

If $\mathcal{P} = \{p_n\}_{n \in \mathbb{N}}$ is countable, then the topology induced by $\mathcal{P}$ is metrizable, with one such metric given by

$$\operatorname{dist}(x, y) = \sum_{n=1}^{\infty} \min(p_n(x - y), 2^{-n})$$

Exercise. Check that this metric does induce the topology induced by $\mathcal{P}$.

Remark. In general, there is no norm on X that induces the same topology. If $\mathcal{P}$ is uncountable, then the induced topology may not be metrizable.

With this in place, we can now look at examples of topologies that we can define on a normed space $(X, ||\cdot||)$.

Definition Strong topology or norm topology
The strong topology τ_s on X is the topology induced by $\mathcal{P} = \{||\cdot||\}$.

Definition Weak topology
The weak topology τ_w on X is the topology induced by the family

$$\mathcal{P} = \{x \mapsto |\langle x, f\rangle| \colon f \in X'\}$$

Definition Weak-$*$ topology
The weak-$*$ topology τ_{w*} on X' is the topology induced by the family

$$\mathcal{P} = \{x \mapsto |\langle x, f\rangle| \colon x \in X\}$$

Recall from II Linear Analysis that X isometrically embeds into X''. In particular, for each $x \in X$, define the map

$$\begin{aligned} \hat{x} \colon X^* &\to \mathbb{R} \\ f &\mapsto f(x) \end{aligned}$$

Note that $\hat{x}$ is linear and bounded: $|\hat{x}(f)| = |f(x)| \leq ||f||||x||$. So $\hat{x} \in X^{**}$ with $||\hat{x}|| \leq ||x||$. Thus, we have

$$\begin{aligned} \hat{\ } : X &\to X^{**} \\ x &\mapsto \hat{x} \end{aligned}$$

This is linear: $\widehat{\lambda x + \mu y}(f) = f(\lambda x + \mu y) = \lambda f(x) + \mu f(y) = (\lambda \hat{x} + \mu \hat{y})(f)$. Moreover, for $x \neq 0$, let $f \in X^*$ be a norming functional at x (exists by Hahn-Banach). Then

$$\hat{x}(f) = f(x) = ||x|| \implies ||\hat{x}|| = ||x||$$

so the canonical map $X \to X^{**}, x \mapsto \hat{x}$ is an isometric embedding into X^{**}.

Remark. Since $\hat{\ }(X) \subset X''$, it is easy to see that the weak topology on X' is coarser than the weak-$*$ topology on X'. If X is reflexive (i.e. $\hat{\ }(X) = X''$), then the two topologies coincide. In that case, we can identify X with X'', so we also have a weak-$*$ topology on X, which coincides with the weak topology on X.

Proposition 3.9 Basic properties
Let X be a Banach space.

(i) *The weak and weak-$*$ topologies are Hausdorff.*
(ii) *The map $x \mapsto \langle x, f\rangle$ is continuous in the weak topology for all $f \in X'$. The map $f \mapsto \langle x, f\rangle$ is continuous in the weak-$*$ topology for all $x \in X$.*
(iii) *A sequence (x_n) in X converges to $x \in X$ in the weak topology iff $\langle x_n, f\rangle \to \langle x, f\rangle$ for all $f \in X'$. We then write $x_n \rightharpoonup x$.*
(iv) *If $x_n \to x$ (in the strong topology), then $x_n \rightharpoonup x$.*
(v) *A sequence (f_n) in X' converges to $f \in X'$ in the weak-$*$ topology if $\langle x, f_n\rangle \to \langle x, f\rangle$ for all $x \in X$. We then write $f_n \overset{*}{\rightharpoonup} f$.*

(vi) *If* $f_n \to f$*, then* $f_n \overset{*}{\rightharpoonup} f$.
(vii) *If* $x_n \rightharpoonup x$ *in* X*, then* $||x_n||$ *is bounded and* $\liminf_{n\to\infty} ||x_n|| \geq ||x||$.
(viii) *If* $f_n \overset{*}{\rightharpoonup} f$ *in* X'*, then* $||f_n||$ *is bounded and* $\liminf_{n\to\infty} ||f_n|| \geq ||f||$.
(ix) *If either* $x_n \rightharpoonup x, f_n \to f$ *or* $x_n \to x, f_n \overset{*}{\rightharpoonup} f$*, then* $\langle x_n, f_n\rangle \to \langle x, f\rangle$.

Remarks

- The weak and weak-$*$ topologies are the coarsest such that $x \mapsto \langle x, f\rangle$ and $f \mapsto \langle x, f\rangle$, respectively, are continuous.
- Property (v) above says that a sequence of bounded linear functionals converge in the weak-$*$ topology iff they converge pointwise on X. By linearity, this is equivalent to pointwise convergence on B_X. In contrast, convergence in the strong topology of X' is equivalent to uniform convergence on B_X.

Proof of Proposition 3.9. We've already proven (i).

(ii) Fix $U \subset \mathbb{R}$ open and $x \in f^{-1}(U)$. Since U is open, $(f(x)-\varepsilon, f(x)+\varepsilon) \subset U$ for some $\varepsilon > 0$. Then $N(x; f, \varepsilon) \subset f^{-1}(U)$. Thus $f^{-1}(U)$ is open in the weak topology.
A similar argument works for the weak-$*$ case.

(iii) ($\Longrightarrow$) Follows from (ii). ($\Longleftarrow$) Fix $f_1, \cdots, f_k$ and $\varepsilon > 0$. Since $\langle x_n, f_i\rangle \to \langle x, f_i\rangle$ for all $i \in \{1, \cdots, k\}$, we can pick $N \in \mathbb{N}$ such that $|\langle x_n - x, f_i\rangle| < \varepsilon$ for all $n \geq N$ and all $i \in \{1, \cdots, k\}$. It follows that $\in N(x; f_1, \cdots, f_k, \varepsilon)$

(iv) Suppose $x_n \to x$. For any $f \in X'$, we have

$$|\langle x_n - x, f\rangle| \leq ||x_n - x||||f|| \to 0 \Longrightarrow \langle x_n, f\rangle \to \langle x, f\rangle$$

so $x_n \rightharpoonup x$.

(v) Similar argument as in (iii).

(vi) Similar argument as in (iv).

(vii) Suppose $x_n \rightharpoonup x$ in X. By (iii), we have

$$\hat{x_n}(f) = \langle x_n, f\rangle \to \langle x, f\rangle = \hat{x}(f) \qquad \forall\, f \in X'$$

Since $\hat{x_n}$ converges pointwise to $\hat{x}$ on X', $\{\hat{x_n}\}$ is pointwise bounded. By the principle of uniform boundedness, it follows that $||x_n|| = ||\hat{x_n}||$ is bounded.
Now, by Hahn-Banach, we can pick $f \in B_{X'}$ a norming functional at $x \in X$. Then

$$||x|| = |\langle x, f\rangle| = \lim_{n\to\infty} |\langle x_n, f\rangle| \leq \liminf_{n\to\infty} ||x_n||$$

(viii) Note that $\{f_n\}$ is pointwise bounded since (f_n) converges pointwise to f. By the principle of uniform boundedness, it follows that $||f_n||$ is bounded. Moreover, for each $x \in B_X$, we have

$$|\langle x, f\rangle| = \lim_{n\to\infty} |\langle x, f_n\rangle| \leq \liminf_{n\to\infty} ||f_n||$$

Taking the supremum over all $x \in B_X$, we deduce that $||f|| \leq \liminf_{n\to\infty} ||f_n||$.

(ix) Suppose $x_n \rightharpoonup x$ and $f_n \to f$. Then

$$\begin{aligned}|\langle x_n, f\rangle - \langle x, f\rangle| &\leq |\langle x_n - x, f\rangle| + |\langle x_n, f_n - f\rangle| \\ &\leq |\langle x - n - x, f\rangle| + ||f_n - f|| \cdot \sup_n ||x_n|| \\ &\to 0\end{aligned}$$

A similar argument works for the other case. ■

Example

- Let E be a compact metric space. Let (μ_n) be a sequence of complex Borel measures on E, and μ another complex Borel measure on E. Then $\mu_n \overset{*}{\rightharpoonup} \mu$ iff
$$\int f\,d\mu_n \to \int f\,d\mu \quad \forall\, f \in C(E)$$
- Let $p, q \in [1, \infty]$ with $p^{-1} + q^{-1} = 1$ and $p > 1$. Let (f_n) be a sequence in $L^p(\mathbb{R}^d)$ and $f \in L^p(\mathbb{R}^d)$. Then $f_n \overset{*}{\rightharpoonup} f$ iff
$$\int f_n g\,dx \to \int fg\,dx \quad \forall\, g \in L^q(\mathbb{R}^d)$$
- Let $\mathcal{H}$ be a Hilbert space. Suppose (x_n) is a sequence in $\mathcal{H}$ and $x \in \mathcal{H}$. Then $x_n \rightharpoonup x$ iff
$$\langle x_n, y\rangle \to \langle x, y\rangle \quad \forall\, y \in \mathcal{H}$$

Example

The following sequences in $L^2(\mathbb{R})$ weakly converge to 0 but do not converge in the strong topology.

- $f_n(x) = e^{inx}\mathbb{1}_{[0,1]}$
- $f_n(x) = 2^{n/2}\mathbb{1}_{[0,2^{-n}]}$
- $f_n(x) = \mathbb{1}_{[n,n+1]}$

Theorem 3.10

Let $\mathcal{H}$ be a Hilbert space. Suppose (x_n) is a sequence in $\mathcal{H}$ with $x_n \rightharpoonup x$ for some $x \in \mathcal{H}$. Then (x_n) converges in the strong topology iff $\limsup_{n\to\infty} ||x_n|| = ||x||$. In this case, the limit in the strong topology is also x.

Proof. See Example Sheet 2. ■

Theorem 3.11 Banach–Alaoglu theorem

Let X be a Banach space. Then $B_{X'}$ is compact in the weak-$*$ topology.

Remark. The proof of the general result relies on Tychonoff's theorem (for infinite products). However, we will only be proving the case where X is separable. After all, this special case suffices for most applications in, for instance, PDE analysis.

Lemma 3.12

Let X be a separable Banach space. If $(x_n) \subset B_X$ is a countable dense subset of B_X, then
$$\operatorname{dist}(f, g) = \sum_{n=1}^{\infty} \frac{|\langle x_n, f - g\rangle|}{2^n}$$
defines a metric on $B_{X'}$, which induces the restriction of the weak-$*$ topology to $B_{X'}$.

Proof. See Example Sheet 2. ■

By the above lemma, the weak-$*$ topology on $B_{X'}$ is metrizable, so compactness is equivalent to sequential compactness. Thus, in the separable case, the Banach-Alaoglu reduces to the following.

Theorem 3.13 Banach–Alaoglu theorem: separable version

Let X be a separable Banach space. Every sequence $(f_n)_{n\in\mathbb{N}}$ in $B_{X'}$ has a weak-$*$ convergent subsequence $(f_{n_k})_{k\in\mathbb{N}}$.

Proof. Since X is separable, we can pick a countable dense subset (x_n) of B_X. By Bolzano–Weierstrass, we can pick $N_1 \subset \mathbb{N}$ such that $(\langle x_1, f_n\rangle)_{n\in N_1}$ converges to some $\alpha_1 \in \triangle$. Proceed inductively to obtain a subset $N_m \subset N_{m-1}$ such that $(\langle x_m, f_n\rangle)_{n\in N_m} \to \alpha_m$ for some $\alpha_m \in \triangle$. Then define $(n_k)_{k\in\mathbb{N}}$ by picking $n_1 < n_2 < n_3 < \cdots$ such that $n_k \in N_k$. By construction, we have that $\lim_{k\to\infty}\langle x_m, f_{n_k}\rangle = \alpha_m$ for all m.

Now, fix $x \in B_X$ and $\varepsilon > 0$. Pick x_m such that $||x - x_m|| < \varepsilon/3$. Moreover, we can pick $K \in \mathbb{N}$ such that $|\langle x_m, f_{n_k} - f_{n_l}\rangle| < \varepsilon/3$ for all $k, l \geq K$. Thus, for all $k, l \geq K$, we have

$$|\langle x, f_{n_k} - f_{n_l}\rangle| \leq 2||x - x_m|| + |\langle x_m, f_{n_k} - f_{n_l}\rangle| < \varepsilon$$

so $(\langle x, f_{n_k}\rangle)_{k\in\mathbb{N}}$ is Cauchy and hence convergent. By homogeneity, this convergence holds for all $x \in X$. Finally, the limit $\lim_{k\to\infty}\langle x, f_{n_k}\rangle$ is in $B_{X'}$ by the uniform boundedness principle. ■

4 Fourier analysis

4.1 The Fourier transform in L^1

Definition Fourier transform
The Fourier transform of a function $f \in L^1(\mathbb{R}^d)$ is defined as

$$\widehat{f}(\xi) = \int f(x) e^{-2\pi i \langle x, \xi \rangle} \, dx$$

Notation. We will sometimes also denote the Fourier transform of f by $\mathcal{F}[f]$.

Proposition 4.1 Basic properties
Let $f, g \in L^1(\mathbb{R}^d)$. Then

(i) $\widehat{f} \in C(\mathbb{R}^d) \cap L^\infty(\mathbb{R}^d)$ *and* $||\widehat{f}||_\infty \leq ||f||_1$
(ii) $\widehat{\lambda f} = \lambda \widehat{f}$ *for all* $\lambda \in \mathbb{C}$
(iii) $\widehat{f+g} = \widehat{f} + \widehat{g}$
(iv) $\widehat{\tau_a f} = e^{-2\pi i \langle a, \xi \rangle} \hat{f}$ *for all* $a \in \mathbb{R}^d$
(v) $\widehat{f \circ U} = |\det U|^{-1} \widehat{f} \circ (U^{-1})^{\mathrm{tr}}$ *for all* $U \in \mathrm{GL}_d(\mathbb{R})$
(vi) $\widehat{\overline{\widehat{f}}} = \mathcal{R}[\overline{\widehat{f}}]$, *where* $\mathcal{R}[f](x) = f(-x)$
(vii) $\widehat{f \star g} = \widehat{f} \cdot \widehat{g}$

Proof. (i): Observe that $\widehat{f} \in L^\infty$ with $||\widehat{f}||_\infty \leq ||f||_1$ since

$$|\widehat{f}(\xi)| \leq \int |f(x) e^{2\pi i \langle x, \xi \rangle}| \, dx = ||f||_1$$

Moreover, note that $|f(x)e^{2\pi i \langle x, \xi \rangle}| \leq |f(x)|$ with $|f| \in L^1$ and that the integrand is continuous as a function of ξ. Thus, $\widehat{f} \in C(\mathbb{R}^d)$ by continuity under the integral sign.
(ii) and (iii) follow from linearity of integration.
(iv): We have

$$\begin{aligned}
\widehat{\tau_a f}(\xi) &= \int f(x-a) e^{-2\pi i \langle x, \xi \rangle} \, dx \\
&= e^{-2\pi i \langle a, \xi \rangle} \int f(x-a) e^{-2\pi i \langle x-a, \xi \rangle} \, dx \\
&= e^{-2\pi i \langle a, \xi \rangle} \int f(x) e^{-2\pi i \langle x, \xi \rangle} \, dx \\
&= e^{-2\pi i \langle a, \xi \rangle} \widehat{f}(\xi)
\end{aligned}$$

(v): Observe that

$$\begin{aligned}
\widehat{f \circ U}(\xi) &= \int (f \circ U)(x) e^{-2\pi i \langle x, \xi \rangle} \, dx \\
&= \int f(Ux) e^{-2\pi i \langle Ux, (U^{-1})^{\mathrm{tr}} \xi \rangle} \, dx \\
&= \int f(y) e^{-2\pi i \langle y, (U^{-1})^{\mathrm{tr}} \xi \rangle} |\det U|^{-1} \, dy \\
&= |\det U|^{-1} \widehat{f} \circ (U^{-1})^{\mathrm{tr}}
\end{aligned}$$

(vi): Note that

$$\widehat{\overline{f}}(\xi) = \int \overline{f(x)} e^{-2\pi i \langle x, \xi \rangle} \, dx = \int \overline{f(x) e^{-2\pi i \langle x, -\xi \rangle}} \, dx = \overline{\hat{f}(-\xi)}$$

(vii): Observe that

$$\iint |f(t)g(x-t)|\,dx\,dt = \int |f(t)|||\tau_t g||_1\,dt = ||f||_1||g||_1 < \infty$$

so we can use Fubini's theorem to obtain

$$\begin{aligned}\widehat{f\star g}(\xi) &= \int \left[\int f(t)g(x-t)\,dt\right] e^{-2\pi i\langle x,\xi\rangle}\,dx \\ &= \iint f(t)g(x-t)e^{-2\pi i\langle x,\xi\rangle}\,dx\,dt \\ &= \int f(t)e^{-2\pi i\langle t,\xi\rangle}\left[\int g(x-t)e^{-2\pi i\langle x-t,\xi\rangle}\,dx\right]\,dt \\ &= \int f(t)e^{-2\pi i\lambda x,\xi\rangle}\widehat{g}(\xi)\,dt \\ &= \widehat{f}(\xi)\widehat{g}(\xi)\end{aligned}$$

as required. ■

Proposition 4.2
Let $f \in \mathcal{S}(\mathbb{R}^d)$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$. Then

$$\widehat{D^\alpha f}(\xi) = (2\pi i)^{|\alpha|}\xi^\alpha \widehat{f}(\xi)$$

Proof. We first prove the result for $|\alpha| = 1$. Fix $j \in \{1, \cdots, d\}$. Since f is smooth, we can use integration by parts from Riemann integration theory to deduce that, for all $a, b \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$\int_a^b \frac{\partial f}{\partial x_j}(x)e^{-2\pi i\langle x,\xi\rangle}\,dx_j = f(x_{(b)})e_{b,x} - f(x_{(a)})e_{a,x} + 2\pi i\xi_j\int_a^b f(x)e^{-2\pi i\langle x,\xi\rangle}\,dx_j$$

where $e_{b,x}, e_{a,x} \in \partial\triangle$ and $x_{(b)} = (x_1, \ldots, b, \ldots, x_d)$. Taking $a \to -\infty$ and $b \to \infty$, we get

$$\int_{\mathbb{R}} \frac{\partial f}{\partial x_j}(x)e^{-2\pi i\langle x,\xi\rangle}\,dx_j = 2\pi i\xi_j\int_{\mathbb{R}} f(x)e^{-2\pi i\langle x,\xi\rangle}\,dx_j$$

where we have used the fact that $f \in \mathcal{S}(\mathbb{R}^d)$ to deduce that the boundary terms vanish and that the integrals converge by dominated convergence theorem. Finally, we integrate over the remaining variables and use Fubini to conclude that, for any $f \in \mathcal{S}(\mathbb{R}^d)$,

$$\widehat{\partial_j f}(\xi) = 2\pi i\xi_j\widehat{f}(\xi)$$

We then proceed by induction. The case $|\alpha| = 0$ is trivial. Suppose the result holds for all $|\alpha| = k$. Since $D^\alpha f \in \mathcal{S}(\mathbb{R}^d)$, it follows that

$$\widehat{D^{\alpha+e_j}f}(\xi) = \partial_j\widehat{(D^\alpha f)}(\xi) = 2\pi i\xi_j\widehat{D^\alpha f}(\xi) = 2\pi i\xi_j[(2\pi i)^{|\alpha|}\xi^\alpha\widehat{f}(\xi)] = (2\pi i)^{|\alpha+e_j|}\xi^{\alpha+e_j}\widehat{f}(\xi)$$

so the result also holds for all $|\alpha| = k+1$, as required. ■

Proposition 4.3
Let $f \in \mathcal{S}(\mathbb{R}^d)$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$. Then $\widehat{f} \in C^\infty$ with

$$D^\alpha\widehat{f} = \mathcal{F}\left[(-2\pi i)^{|x|}x^\alpha f\right]$$

where we use $\mathcal{F}$ as an alternative notation for the Fourier transform.

Proof. We proceed by induction on $|\alpha|$. The base case ($|\alpha| = 0$) is trivial. Suppose the result

holds for $|\alpha| = k$. By the mean value theorem,

$$\frac{|(-2\pi i)^{|\alpha|}x^\alpha f(x)||e^{-2\pi i\langle x,\xi\rangle}||e^{-2\pi i x_j h} - 1|}{|h|} \leq (2\pi)^{|\alpha|+1}|x^{\alpha+e_j}||f(x)|$$

with $(2\pi)^{|\alpha|+1}|x^{\alpha+e_j}||f(x)| \in L^1(\mathbb{R}^d)$ by decay properties of f as a Schwartz function. We can then differentiate under the integral sign to obtain

$$\begin{aligned} D^{\alpha+e_j}\widehat{f}(\xi) &= \frac{\partial}{\partial \xi_j}\int_{\mathbb{R}^d}(-2\pi i)^{|\alpha|}x^\alpha f(x)e^{-2\pi i\langle x,\xi\rangle}\,dx \\ &= \int_{\mathbb{R}^d}(-2\pi i)^{|\alpha|}x^\alpha f(x)\frac{\partial}{\partial \xi_j}e^{-2\pi i\langle x,\xi\rangle}\,dx \\ &= \int_{\mathbb{R}^d}(-2\pi i)^{|\alpha+e_j|}x^{\alpha+e_j}f(x)\,dx \end{aligned}$$

as required. ■

Remark. The above proposition tells us that every partial derivative of the Fourier transform of a Schwartz function is the Fourier transform of a Schwartz function.

Proposition 4.4
If $f \in \mathcal{S}(\mathbb{R}^d)$, then $\widehat{f} \in \mathcal{S}(\mathbb{R}^d)$.

Proof. We have already seen that $\widehat{f} \in C^\infty(\mathbb{R}^d)$. It thus remains to show that $\sup_\xi ||\xi|^n D^\alpha \widehat{f}(\xi)| < \infty$ for all $\alpha \in \mathbb{Z}^d_{\geq 0}$ and $n \in \mathbb{Z}_{\geq 0}$. By the above remark, it suffices to prove the claim for $\alpha = 0$.

By Proposition 4.2, we have that

$$\widehat{\Delta f}(\xi) = -4\pi^2|\xi|^2\widehat{f}(\xi)$$

Iterating, we get

$$\widehat{\Delta^n f}(\xi) = (-4\pi^2)^n|\xi|^{2n}\widehat{f}(\xi)$$

By Proposition 4.1, we then have that

$$\sup_\xi \left||\xi|^{2n}\widehat{f}(\xi)\right| = \frac{1}{(4\pi^2)^n}||\widehat{\Delta^n f}||_\infty \leq \frac{1}{(4\pi^2)^n}||\Delta^n f||_1 < \infty$$

It then follows that, for odd n, we also have

$$\sup_\xi \left||\xi|^n\widehat{f}(\xi)\right| \leq \max\left\{\sup_{|\xi|\leq 1}|\widehat{f}(\xi)|, \sup_{|\xi|\geq 1}\left||\xi|^{n+1}\widehat{f}(\xi)\right|\right\} < \infty$$

as required. ■

Remarks

- Observe from the previous results that there is a duality between smoothness and decay properties of a function and its Fourier transform.
- The assumption $f \in \mathcal{S}(\mathbb{R}^d)$ in the two propositions above is overkill. We will return to this point later.

Theorem 4.5 Riemann-Lebesgue lemma
Let $f \in L^1(\mathbb{R}^d)$. Then $\widehat{f}(\xi) \to 0$ as $|\xi| \to \infty$.

Proof. Fix $\varepsilon > 0$. By density, we can pick $g \in C_c^\infty(\mathbb{R}^d)$ such that $||f - g||_1 < \varepsilon$. Note that $g \in \mathcal{S}(\mathbb{R}^d)$, so we also have that $\widehat{g} \in \mathcal{S}(\mathbb{R}^d)$ and thus $\widehat{g}(\xi) \to 0$ as $|\xi| \to \infty$. By Proposition 4.1,

$||\widehat{f-g}||_\infty \leq ||f-g||_1 < \varepsilon$. We then have that

$$\limsup_{|\xi|\to\infty} |\widehat{f}(\xi)| \leq \limsup_{|\xi|\to\infty} |(\widehat{f-g})(\xi)| + \limsup_{|\xi|\to\infty} |\widehat{h}(\xi)| < \varepsilon$$

as required. ■

Theorem 4.6 Fourier inversion theorem
Let $f \in L^1(\mathbb{R}^d)$ be such that $\widehat{f} \in L^1(\mathbb{R}^d)$. Then

$$f(x) = \int_{\mathbb{R}^d} \widehat{f}(\xi) e^{2\pi i \langle x,\xi\rangle}\, d\xi$$

for a.e. $x \in \mathbb{R}^d$.

Remarks

- Of course, the above theorem also implies that $f \in L^1(\mathbb{R}^d)$ has a continuous representative for which equality holds for all $x \in \mathbb{R}^d$.
- To prove the theorem, a physicist might try to write

$$\int_{\mathbb{R}^d} \widehat{f}(\xi) e^{2\pi i \langle x,\xi\rangle}\, d\xi = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} f(y) e^{-2\pi i\langle y,\xi\rangle} e^{2\pi i\langle x,\xi\rangle}\, dy\, d\xi$$

  and proceed by changing the order of integration and recognising the integral $\int e^{2\pi i\langle x-y,\xi\rangle}\, d\xi$ as $\delta(x-y)$. As analysts, we shall not engage in such jiggery-pokery. Nonetheless, heuristics involving delta functions can often be made rigorous by mollification

Proof of Theorem 4.6. Note that, for any $g \in L^1(\mathbb{R}^d)$, we have

$$\begin{aligned}
\int \widehat{f}(\xi) e^{2\pi i\langle x,\xi\rangle} g(\xi)\, d\xi &= \iint f(x) e^{2\pi i\langle x-y,\xi\rangle} g(\xi)\, dy\, d\xi \\
&= \iint f(y) e^{2\pi i\langle x-y,\xi\rangle} g(\xi)\, d\xi\, dy \\
&= \int f(y)\widehat{g}(y-x)\, dy \\
&= (f \star \mathcal{R}[\widehat{g}])(x)
\end{aligned}$$

Now, it suffices to pick a sequence $(g_n)_{n\in\mathbb{N}}$ in $L^1(\mathbb{R}^d)$ such that $||g_n||_\infty \leq 1$ for all $n \in \mathbb{N}$, $g_n(\xi) \to 1$ for a.e. $\xi \in \mathbb{R}^d$, and $(\mathcal{R}[\widehat{g_n}])$ is an approximation of the identity. This is because, by dominated convergence theorem, we have that

$$\int \widehat{f}(\xi) e^{2\pi i\langle x,\xi\rangle}\, d\xi = \lim_{n\to\infty} \int \widehat{f}(\xi) e^{2\pi i\langle x,\xi\rangle} g_n(\xi)\, d\xi = \lim_{n\to\infty} (f \star \mathcal{R}[\widehat{g_n}])(x)$$

By Theorem 2.12, we then have that $f \star \mathcal{R}[\widehat{g_n}] \to f$ in L^1. By passing to a subsequence, $f \star \mathcal{R}[\widehat{g_n}] \to f$ a.e.

It remains to show that such a sequence exists. A concrete example is given by

$$g_n(\xi) = e^{-\pi(\xi/n)^2}, \qquad \widehat{g_n}(x) = n^d e^{-\pi(nx)^2}$$

It is easy to verify that this has the required properties. ■

Theorem 4.7 Plancherel's formula for Schwartz functions
Let $f, g \in \mathcal{S}(\mathbb{R}^d)$. Then

$$\int f(x)\overline{g(x)}\, dx = \int \widehat{f(\xi)}\overline{\widehat{g}(\xi)}\, d\xi$$

Proof. By Proposition 4.1, $\widehat{f \star \mathcal{R}[\overline{g}]} = \widehat{f} \cdot \widehat{\mathcal{R}[\overline{g}]} = \widehat{f} \cdot \overline{\widehat{g}}$. By Theorem 4.6, it follows that

$$\int f(x)\overline{g(x)}\,dx = (f \star \mathcal{R}[\overline{g}])(0) = \int \widehat{f} \cdot \overline{\widehat{g}(\xi)}\,d\xi$$

as required. ■

4.2 The Fourier transform in L^2

Theorem 4.7 implies that the map $f \mapsto \widehat{f}$ is a linear transformation on $\mathcal{S}(\mathbb{R}^d)$ that preserves the inner product defining the Hilbert space $L^2(\mathbb{R}^d)$. Since $\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$, we can extend our definition of the Fourier transform to $f \in L^2(\mathbb{R}^d)$. Concretely, pick (f_n) in $\mathcal{S}(\mathbb{R}^d)$ such that $f_n \to f$ in L^2. Then (f_n) is Cauchy. By Plancherel's formula (Theorem 4.7), $(\widehat{f_n})$ is also a Cauchy sequence. We then set $\widehat{f} := \lim_{n\to\infty} \widehat{f_n}$ (in L^2).

Remark. It is easy to see that $\widehat{f}$ is well-defined.

- If (g_n) is another such approximating sequence, then $||g_n - f_n||_2 \to 0$ implies that $||\widehat{g_n - f_n}||_2 \to 0$ so $\lim \widehat{g_n} = \lim \widehat{f_n}$.
- Suppose $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. Pick (f_n) in $\mathcal{S}(\mathbb{R}^d)$ such that $f_n \to f$ in L^1 and L^2. Of course, $\widehat{f_n}$ pointwise converges to the L^1 definition of $\widehat{f}$ and converges in L^2 to the L^2 definition of $\widehat{f}$. Passing to a subsequence, L^2 convergence becomes a.e. convergence. By uniqueness of pointwise limits, the two definitions of $\widehat{f}$ agree a.e. on $\mathbb{R}^d$.

Proposition 4.8 Basic properties
Let $f, g \in L^2(\mathbb{R}^d)$.

(i) $\widehat{f} \in L^2(\mathbb{R}^d)$ *and* $||\widehat{f}||_2 = ||f||_2$
(ii) $\langle f, g\rangle = \langle \widehat{f}, \widehat{g}\rangle$
(iii) $\widehat{\widehat{f}} = \mathcal{R}[f]$
(iv) $\widehat{f \cdot g} = \widehat{f} \star \widehat{g}$

Proof. The proof of these properties follow from approximation by Schwartz functions. We will prove (iv) in detail as illustration; the rest are left as an exercise.

STEP 1: Schwartz functions
For $f, g \in \mathcal{S}(\mathbb{R}^d)$, we can use basic properties (Proposition 4.1) to deduce that

$$\widehat{\widehat{f} \star \widehat{g}} = \widehat{\widehat{f}} \cdot \widehat{\widehat{g}} = \mathcal{R}[f] \cdot \mathcal{R}[g]$$

Applying Fourier inversion (Theorem 4.6), we get $\widehat{f} \star \widehat{g} = \widehat{f \cdot g}$.

STEP 2: Extension to L^2
Since $\mathcal{S}(\mathbb{R}^d)$ is dense in $L^2(\mathbb{R}^d)$, we can pick sequences (f_n) and (g_n) in $\mathcal{S}(\mathbb{R}^d)$ such that $f_n \to f$ and $g_n \to g$ in $L^2(\mathbb{R}^d)$. By Step 1, the result holds for every pair (f_n, g_n). We have $\widehat{f_n} \star \widehat{g_n} \to \widehat{f} \star \widehat{g}$ in $L^\infty(\mathbb{R}^d)$ since, by Young's inequality (Theorem 2.7), we can estimate

$$\begin{aligned} ||\widehat{f_n} \star \widehat{g_n} - \widehat{f} \star \widehat{g}||_\infty &\le ||\widehat{f_n} \star \widehat{g_n} - \widehat{f_n} \star \widehat{g}||_\infty + ||\widehat{f_n} \star \widehat{g_n} - \widehat{f} \star \widehat{g}||_\infty \\ &\le ||\widehat{f_n}||_{L^2}||\widehat{g_n} - \widehat{g}||_{L^2} + ||\widehat{f} - \widehat{f_n}||_{L^2}||\widehat{g}||_{L^2} \to 0 \end{aligned}$$

Moreover, we have $f_n \cdot g_n \to f \cdot g$ in $L^1(\mathbb{R}^d)$ since, by Cauchy-Schwarz, we can estimate

$$\begin{aligned} ||f_n g_n - fg||_1 &\le ||(f_n - f)g||_1 + ||f_n(g - g)||_1 \\ &\le ||f_n - f||_2||g||_2 + ||f_n||_2||g_n - g||_2 \to 0 \end{aligned}$$

It then follows from Proposition 4.1(i) that $\widehat{f_n \cdot g_n} \to \widehat{f \cdot g}$ in $L^\infty(\mathbb{R}^d)$. Hence, $\widehat{f} \star \widehat{g} = \widehat{f \cdot g}$. ■

4.3 Periodic functions

We say that a function $f\colon \mathbb{R}^d \to \mathbb{C}$ is *periodic* if $f(x+a) = f(x)$ for all $a \in \mathbb{Z}^d$ and $x \in \mathbb{R}^d$. For $p \in [1, \infty)$, we define

$$||f||_{L^p(\mathbb{T}^d)} = \left(\int_{[0,1]^d} |f(x)|^p \, dx \right)^{1/p}$$

and

$$||f||_{L^\infty(\mathbb{T}^d)} = \text{ess sup} \, |f|$$

For $p \in [1, \infty]$, we then denote by $L^p(\mathbb{T}^d)$ the space of measurable functions for which $||f||_p < \infty$, with functions identified if they agree a.e.

Remark. A remark is in order regarding the notation used. Here, $\mathbb{T}^d$ stands for $\mathbb{R}^d/\mathbb{Z}^d$, which may be endowed with the structure of a manifold and a group. In fact, there is a unique probability measure on it that is invariant under translations with respect to the group operation. Periodic functions can then be identified with functions on $\mathbb{R}^d/\mathbb{Z}^d$, and we can talk about L^p spaces with respect to the above measure. The resulting norm is then equivalent to $|| \cdot ||_{L^\infty(\mathbb{T}^d)}$. However, this more sophisticated viewpoint is not necessary for this course.

Definition Fourier coefficients
The Fourier coefficients of $f \in L^1(\mathbb{T}^d)$ are defined by

$$\widehat{f}(\xi) = \int_{[0,1]^d} f(x) e^{-2\pi i \langle x, \xi \rangle} \, dx$$

for $\xi \in \mathbb{Z}^d$.

> Unlike for functions on $\mathbb{R}^d$, the functions $(x \mapsto e^{2\pi i \langle x, \xi \rangle})_{\xi \in \mathbb{Z}^d}$ form an orthonormal basis in $L^2(\mathbb{T}^d)$, and the Fourier coefficients of the basis elements when we decompose f.

Now, let us explore the relationship between the Fourier coefficients above and the Fourier transform from the previous subsections. Consider the map $P\colon L^1(\mathbb{R}^d) \to L^1(\mathbb{T}^d)$ defined by

$$P[f](x) = \sum_{a \in \mathbb{Z}^d} f(x-a) = \sum_{a \in \mathbb{Z}^d} \tau_a f(x)$$

This is well-defined: it is clearly periodic and we can estimate $||Pf||_{L^1(\mathbb{T}^d)} \leq ||f||_{L^1(\mathbb{R}^d)}$ since $|Pf(x)| \leq P|f|(x)$ and

$$\int_{\mathbb{T}^d} Pf \, dx = \int_{[0,1]^d} \sum_{a \in \mathbb{Z}^d} \tau_a f \, dx = \sum_{a \in \mathbb{Z}^d} \int_{[0,1]^d} \tau_a f \, dx = \sum_{a \in \mathbb{Z}^d} \int_{[0,1]^d + a} f \, dx = \int_{\mathbb{R}^d} f \, dx$$

Above, Fubini was used to exchange the sum and integral, which can be justified by showing that $\int_{\mathbb{T}^d} \sum_{a \in \mathbb{Z}^d} |\tau_a f| \, dx < \infty$.

The map P has a partial inverse. To see why, pick $u \in L^1(\mathbb{R}^d)$ such that $P[u] \equiv 1$. A concrete example is given by $u = \mathbb{1}_{[-1/2, 1/2)^d}$ but we will later see that u can be chosen to be smooth.

Note that, for every $\varphi \in L^1(\mathbb{T}^d)$, we have

$$P[u\varphi](x) = \sum_{a\in\mathbb{Z}^d} u(x-a)\varphi(x-a) = \varphi(x)\cdot P[u](x) = \varphi(x)$$

Thus, P is surjective. However, $uP[f]$ differs from f in general, so P is not injective. Note also that, for every $\varphi \in L^1(\mathbb{T}^d)$, we have

$$\int_{\mathbb{T}^d} \varphi\, dx = \int_{\mathbb{T}^d} P[u\varphi]\, dx = \int_{\mathbb{R}^d} u\varphi\, dx$$

From the above computations, we obtain the following relationship between the Fourier transform on $L^1(\mathbb{R}^d)$ and the Fourier coefficients on $L^1(\mathbb{T}^d)$:

$$\begin{aligned} \widehat{Pf}(\xi) &= \widehat{f}(\xi), && \text{for } f \in L^1(\mathbb{R}^d) \text{ and } \xi \in \mathbb{Z}^d \\ \widehat{\varphi}(\xi) &= \widehat{u\varphi}(\xi), && \text{for } \varphi \in L^1(\mathbb{T}^d) \text{ and } \xi \in \mathbb{Z}^d \end{aligned}$$

Theorem 4.9
Let $\varphi \in L^1(\mathbb{T}^d)$. If $\sum_{\xi\in\mathbb{Z}^d} |\widehat{\varphi}(\xi)| < \infty$, then for a.e. $x \in \mathbb{T}^d$,

$$\varphi(x) = \sum_{\xi\in\mathbb{Z}^d} \widehat{\varphi}(\xi) e^{2\pi i\langle x,\xi\rangle}$$

Remark. This is analogous to the Fourier inversion theorem for $L^1(\mathbb{R}^d)$. As in that theorem, $f \in L^1(\mathbb{R}^d)$ has a continuous representative for which equality holds for all $x \in \mathbb{T}^d$.

Proof. Fix $a_n(\xi) \in [0,1]$ for each $n \in \mathbb{N}$ and $\xi \in \mathbb{Z}^d$ such that $\lim_{n\to\infty} a_n(\xi) = 1$. Then

$$\sum_{\xi\in\mathbb{Z}^d} \widehat{\varphi}(\xi) e^{2\pi i\langle x,\xi\rangle} = \lim_{n\to\infty} \sum_{\xi\in\mathbb{Z}^d} \widehat{\varphi}(\xi) e^{2\pi i\langle x,\xi\rangle} a_n(\xi)$$

by dominated convergence. Observe that, for $u \in \mathbb{1}_{[-1/2,1/2)^d}$,

$$\sum_{\xi\in\mathbb{Z}^d} \int_{\mathbb{T}^d} \varphi(y) e^{2\pi i\langle x-y,\xi\rangle} a_n(\xi)\, dy = \sum_{\xi\in\mathbb{Z}^d} \int_{\mathbb{R}^d} u\varphi(y) e^{2\pi i\langle x-y,\xi\rangle}\, dy = [(u\varphi)\star g_n](x)$$

where

$$g_n(x) = \sum_{\xi\in\mathbb{Z}^d} e^{2\pi i\langle x,\xi\rangle} a_n(\xi)$$

You will show on Example Sheet 3 that the sequence (a_n) may be chosen such that (ug_n) is an approximation of the identity. By Theorem 2.12, $(u\varphi)\star(ug_n) \to u\varphi$ in $L^1(\mathbb{R}^d)$. By passing to a subsequence, we get $(u\varphi)\star(ug_n) \to u\varphi$ a.e. Note that $g_n = \sum_{a\in\mathbb{Z}^d} \tau_a(u) g_n = \sum_{a\in\mathbb{Z}^d} \tau_a(ug_n)$, so

$$(u\varphi)\star g_n = \sum_{a\in\mathbb{Z}^d} (u\varphi)\star(\tau_a(ug_n))$$

Note that $\operatorname{supp}[(u\varphi)\star(\tau_a(ug_n))] \subset a + [-1,1]^d$, so for any given $x \in \mathbb{R}^d$, $(u\varphi)\star(\tau_a(ug_n))$ vanishes for all but finitely many $a \in \mathbb{Z}^d$. Thus,

$$\sum_{a\in\mathbb{Z}^d} (u\varphi)\star(\tau_a(ug_n)) = \sum_{a\in\mathbb{Z}^d} \tau_a[(u\varphi)\star(ug_n)] \to \sum_{a\in\mathbb{Z}^d} \tau_a(u\varphi) = \varphi$$

almost everywhere on $\mathbb{T}^d$. ■

Theorem 4.10 Poisson summation formula
Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then

$$\sum_{x\in\mathbb{Z}^d} f(x) = \sum_{\xi\in\mathbb{Z}^d} \widehat{f}(\xi)$$

Proof. Note that $|f(x)| \leq C(1+|x|)^{-d-1}$ for a suitable constant $C > 0$. Then $Pf = \sum_{a \in \mathbb{Z}^d} f$ converges uniformly on bounded sets and thus $Pf \in C(\mathbb{T}^d)$ with $Pf(0) = \sum_{x \in \mathbb{Z}^d} f(x)$. By Fourier inversion (Theorem 4.6), we have

$$Pf(0) = \sum_{\xi \in \mathbb{Z}^d} \widehat{Pf}(\xi) = \sum_{\xi \in \mathbb{Z}^d} \widehat{f}(\xi)$$

as required. ■

5 Distribution theory

In this section, we introduce the theory of distributions: a class of objects that include the function spaces we have seen thus far but which are easier to manipulate and allows us to make sense of so-called weak solutions to PDEs. In some sense, this is akin to extending the field $\mathbb{R}$ to the field $\mathbb{C}$, which is algebraically closed.

The idea behind this construction comes from the fact that $L^p(\mathbb{R}^d)' = L^q(\mathbb{R}^d)$, i.e. a function in L^q for $q \in (1, \infty]$ can be viewed as a functional on L^p for the conjugate index $p \in [1, \infty)$. (Note also that L^1 is isometrically embedded into L^∞.) We will be restricting these functionals to a subspace of so-called *test functions*, which would, for instance, allow us to define so-called *weak derivatives* of functions that may otherwise be non-differentiable.

5.1 Test functions and distributions

Fix an open subset $\Omega \subset \mathbb{R}^d$.

Theorem 5.1
There exists a topology on the space $C_c^\infty(\Omega)$ such that

(i) *a sequence (f_n) in $C_c^\infty(\Omega)$ converges to some $f \in C_c^\infty(\Omega)$ iff there exists a compact subset $K \subset \Omega$ such that* $\operatorname{supp} f_n \subset K$ *for all n and $D^\alpha f_n \to D^\alpha f$ uniformly for all $\alpha \in \mathbb{Z}^d_{\geq 0}$*

(ii) *a linear map $A\colon C_c^\infty(\Omega) \to V$ to a topological vector space is continuous iff $A(f_n) \to A(f)$ in V for every convergent sequence (f_n) in $C^\infty(\Omega)$ with $f_n \to f$ in $C_c^\infty(\Omega)$*

Remark. We will not prove this theorem in this course. For those who are interested, a construction is given in Claude Warnick's Analysis of Functions notes.

Definition Space of test functions
The space $\mathcal{D} = \mathcal{D}(\Omega)$ is defined as the space $C_c^\infty(\Omega)$ endowed with a topology from Theorem 5.1. We call its elements test functions.

Definition Space of distributions
The space $\mathcal{D}'(\Omega)$ is defined as the space of continuous linear functionals on $\mathcal{D}(\Omega)$. We call its elements distributions.

Remark. As a dual space, $\mathcal{D}'$ naturally comes with the weak-$*$ topology.

An important characterisation of $\mathcal{D}'$ is given by the following lemma, which we will also not prove.

Lemma 5.2 Continuity of a linear functional
Let $u\colon \mathcal{D}(\Omega) \to \mathbb{C}$ be a linear functional. The following are equivalent:

(i) $u \in \mathcal{D}'(\Omega)$

(ii) $u[\varphi_j] \to u[\varphi]$ *in $\mathbb{C}$ whenever $\varphi_j \to \varphi$ in $\mathcal{D}(\Omega)$*

(iii) *For every compact subset $K \subset \Omega$, there exist constants C and k such that*

$$|u(\varphi)| \leq C||\varphi||_{C^k(K)} \qquad \forall\, \varphi \in \mathcal{D}(\Omega) \text{ with } \operatorname{supp} \varphi \subset K$$

If the property in (iii) holds for u for some k that is independent of K, then we say that u is *of finite order*, and we say that the *order of u* is the smallest such k.

Recall that the local Lebesgue space $L^1_{\text{loc}}(\Omega)$ is defined as

$$L^1_{\text{loc}}(\Omega) := \{f\colon E \to \mathbb{C}\colon f\mathbb{1}_K \in L^1(\Omega)\ \forall\ \text{compact } K \subset \Omega\}$$

Proposition 5.3 Functions as distributions
For each $f \in L^1_{\text{loc}}(\Omega)$, the map

$$\begin{aligned} T_f\colon \mathcal{D} &\to \mathbb{C} \\ g &\mapsto \int_\Omega fg\,dx \end{aligned}$$

is a well-defined element of $\mathcal{D}'$. Moreover, the map

$$\begin{aligned} L^1_{\text{loc}}(\Omega) &\to \mathcal{D}' \\ f &\mapsto T_f \end{aligned}$$

is an injective linear transformation.

Remark. This shows that we can regard functions in $L^1_{\text{loc}}(\mathbb{R}^d)$ as distributions.

Proof. If (g_n) is a sequence in $\mathcal{D}$ with $g_n \to g$ in $\mathcal{D}$, then there exists a compact $K \subset \Omega$ such that $\operatorname{supp} g_n \subset K$ for all n, and $g_n \to g$ uniformly. Therefore,

$$\lim_{n\to\infty} \int_\Omega fg_n\,dx = \lim_{n\to\infty} \int_K fg_n\,dx = \int_K fg\,dx = \int_\Omega fg\,dx$$

by dominated convergence since the restriction of f to K is integrable.

Linearity of $f \mapsto T_f$ is immediate from linearity of integration. Injectivity follows from the following lemma. ■

Lemma 5.4
Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ such that $\varphi(x) \geq 0$ for all $x \in \mathbb{R}^d$ and $\int_{\mathbb{R}^d} \varphi\,dx = 1$. For each $n \in \mathbb{N}$, let $\varphi_n(x) = n^d\varphi(nx)$. Let $K \subset \Omega$ be compact and $f \in L^1_{\text{loc}}(\Omega)$. For each $n \in \mathbb{N}$, define $f_n(x) = T_f[\tau_x\varphi_n]$ for $\{x \in \mathbb{R}^d\colon \operatorname{dist}(x, \partial\Omega) \geq n^{-1}\}$. Then

$$f_n \to f \qquad \text{in } L^1(K)$$

Remark. To be more precise, there exists $N \in \mathbb{N}$ such that $\operatorname{dist}(K, \partial\Omega) > N^{-1}$. Then $(f_n|_K)_{n\geq N}$ is a well-defined sequence in $L^1(K)$. We can then make sense of the convergence of that sequence in $L^1(K)$. In the proof, we will assume that we are working with this sequence, while still writing (f_n) for ease of notation.

Proof. Note that $T_f(\tau_x\check{\varphi}_n) = (f \star \varphi_n)(x)$. By Theorem 2.12, we have $f \star \varphi_n \to f$ in $L^1(K)$. ■

Remark. We now explain why this implies injectivity of $f \mapsto T_f$. Indeed, if $T_f = 0$ in $\mathcal{D}'(\Omega)$, then we must have $f_n = 0$ everywhere on K for all $n \geq N$, so it follows that $||f|_K||_1 = 0$ and thus $f = 0$ a.e. on K. This holds for all compact subsets $K \subset \Omega$, so $f = 0$ in $L^1_{\text{loc}}(\Omega)$.

Proposition 5.5 Measures as distributions
If μ is a Borel measure on Ω such that $\mu(K) < \infty$ for all compact $K \subset \Omega$, then

$$T_\mu(g) = \int_\Omega g\,d\mu$$

defines a distribution. Moreover, if $T_\mu = T_\nu$ in $\mathcal{D}'(\Omega)$, then $\mu = \nu$ as Borel measures on Ω.

Exercise. Prove the above proposition.

Example *Dirac delta*
For any $x \in \Omega$, the Dirac measure δ_x defines a distribution $f \mapsto f(x)$. We will also denote the distribution by δ_x.

Example
Consider the principal value

$$\text{p.v.}\left(\frac{1}{x}\right)(g) := \lim_{r \searrow 0} \int_{\mathbb{R}\setminus[-r,r]} \frac{g(x)}{x}\,dx$$

Note that

$$\begin{aligned}
\int_{[-1,-r]\cup[r,1]} \frac{g(x)}{x}\,dx &= \int_r^1 \frac{g(x)-g(-x)}{x}\,dx \\
&= \int_r^1 \int_{-x}^{x} \frac{g'(t)}{x}\,dt\,dx \\
&= \int_{-1}^1 \int_{\max(|t|,r)}^1 \frac{g'(t)}{x}\,dx\,dt \\
&= \int_{-1}^1 -\log(\max(|t|,r))g'(t)\,dt \\
&\to \int_{-1}^1 -\log|t|g'(t)\,dt
\end{aligned}$$

Thus,

$$\text{p.v.}\left(\frac{1}{x}\right)(g) = \int_{\mathbb{R}\setminus[-1,1]} \frac{g(x)}{x}\,dx - \int_{-1}^1 \log|x|g'(x)\,dx$$

is a well-defined distribution of order at most 1. (It can be shown that it is in fact not of order zero and must thus be of order one.)

5.2 Basic operations on distributions

Definition Multiplication by smooth functions
Let $f \in C^\infty(\Omega)$ and $u \in \mathcal{D}'(\Omega)$. We define $fu \in \mathcal{D}'(\Omega)$ by

$$(fu)[\varphi] := u[h\varphi] \qquad \forall\, \varphi \in \mathcal{D}(\Omega)$$

Exercise. Prove using Lemma 5.2 that fu is a well-defined distribution.

Lemma 5.6
Let $f \in C^\infty(\Omega)$ and $g \in L^1_{\text{loc}}(\Omega)$. Then

$$fT_g = T_{fg}$$

Proof. Observe that

$$fT_g[\varphi] = T_g[f\varphi] = \int f\varphi g\,dx = \int \varphi f g\,dx = T_{fg}[\varphi]$$

for every $\varphi \in \mathcal{D}(\Omega)$. ■

Definition Derivative of a distribution
Let $u \in \mathcal{D}'(\Omega)$. For each $\alpha \in \mathbb{Z}^d_{\geq 0}$, we define $D^\alpha u \in \mathcal{D}'(\Omega)$ by

$$(D^\alpha u)[\varphi] := (-1)^{|\alpha|} u[D^\alpha \varphi]$$

Exercise. Prove using Lemma 5.2 that $D^\alpha u$ is a well-defined distribution.

Remark. If $u \in C_c^\infty(\Omega)$, then its weak derivatives coincide with its classical derivatives. This can be seen by integration by parts.

Lemma 5.7
Let $k \in \mathbb{Z}_{\geq 0}$. Let $f \in C^k(\Omega)$ and $\alpha \in \mathbb{Z}_{\geq 0}^d$ with $|\alpha| \leq k$. Then

$$D^\alpha T_f = T_{D^\alpha f}$$

Proof. Observe that

$$(D^\alpha T_f)[\varphi] = (-1)^{|\alpha|} T_f[D^\alpha \varphi] = (-1)^{|\alpha|} \int f D^\alpha \varphi \, dx = \int \varphi D^\alpha f \, dx = T_{D^\alpha f}[\varphi]$$

for all $\varphi \in \mathcal{D}(\Omega)$. ■

Example *Heaviside function*
The Heaviside function is defined as

$$\begin{aligned} H \colon \Omega = \mathbb{R} &\to \mathbb{R} \\ x &\mapsto \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \end{aligned}$$

We then have

$$\frac{d}{dx} T_H(h) = -\int_0^\infty \frac{dh}{dx} \, dx = h(0)$$

Thus, $dT_H/dx = \delta_0$.

5.3 Convolutions and fundamental solutions

Definition Support of a distribution
Let $u \in \mathcal{D}'(\Omega)$. We define the support of u to be

$$\operatorname{supp} u = \bigcap_{\substack{K \subset \Omega \text{ closed} \\ u(f)=0 \ \forall f \in C_c^\infty(\Omega),\ \operatorname{supp}(f) \cap K = \varnothing}} K$$

Lemma 5.8
Let $u \in \mathcal{D}'(\Omega)$ and $h \in \mathcal{D}(\Omega)$ such that $\operatorname{supp}(h) \cap \operatorname{supp}(u) = \varnothing$. Then $u(h) = 0$.

Proof. Non-examinable. ■

Definition Essential support
The essential support of a measurable function $f \colon \Omega \to \mathbb{C}$ is defined as

$$\operatorname{ess\,supp} f = \bigcap_{\substack{F \text{ closed} \\ f=0 \text{ a.e. on } \Omega \setminus F}} F$$

Lemma 5.9
Let $f \in L^1_{\text{loc}}(\Omega)$. Then $\operatorname{supp} T_f = \operatorname{ess\,supp} f$.

Proof. Suppose that $h \in \mathcal{D}(\Omega)$ such that $\operatorname{supp} h \cap \operatorname{ess\,supp} f = \varnothing$. Then

$$T_f(h) = \int_\Omega f h \, dx = 0$$

Thus, $\operatorname{supp} T_f \subset \operatorname{ess\,supp} f$.

Now, let $K = \operatorname{supp} T_f$ and $\tilde{\Omega} = \Omega \backslash K$. If $h \in \mathcal{D}(\Omega)$ with $\operatorname{supp} h \subset \tilde{\Omega}$, then $T_f(h) = T_{f|_{\tilde{\Omega}}}(h) = 0$. By injectivity of $f \mapsto T_f$ (Lemma 5.3), we have that $f|_{\tilde{\Omega}} = 0$. ■

Definition Convolution of a distribution and a test function
Let $u \in \mathcal{D}'(\mathbb{R}^d)$ and $\psi \in \mathcal{D}(\mathbb{R}^d)$. We define the convolution of u with ψ as

$$\begin{aligned} u \star \psi \colon \mathbb{R}^d &\to \mathbb{C} \\ x &\mapsto u[\tau_x \check{\psi}] \end{aligned}$$

Lemma 5.10 Basic properties
Let $u \in \mathcal{D}'(\mathbb{R}^d)$ and $\psi \in \mathcal{D}(\mathbb{R}^d)$.

(i) *$u \star \psi \in C^\infty(\mathbb{R}^d)$ with*

$$D^\alpha(u \star \psi) = u \star (D^\alpha \psi) = (D^\alpha u) \star \psi$$

(ii) *If, in addition, u has compact support, then $u \star \psi \in \mathcal{D}(\mathbb{R}^d)$ with* $\operatorname{supp}(u \star \psi) \subset \operatorname{supp} u + \operatorname{supp} \psi$

Proof. You will show on Example Sheet 3 that, for any convergent sequence $x_n \to x$ in $\mathbb{R}^d$, we have $\tau_{x_n} \check{\psi} = \tau_x \check{\psi}$ in $\mathcal{D}(\mathbb{R}^d)$.

(i) For $j \in \{1, \cdots, d\}$, we have

$$\frac{(u \star \psi)(x + he_j) - (u \star \psi)(x)}{h} = u\left[\frac{\tau_{x+he_j}\check{\psi} - \tau_x \check{\psi}}{h}\right]$$

for all $h \in [-1, 1]$. On Example Sheet 3, you will show that it follows that

$$\frac{\tau_{x+he_j}\check{\psi} - \tau_x \check{\psi}}{h} \to \tau_x \partial_j \check{\psi} \qquad \text{in } \mathcal{D}(\mathbb{R}^d)$$

as $h \to 0$. By continuity of u, it follows that $\partial_j(u \star \psi) = u \star (\partial_j \psi)$. By iteration, we deduce that $u \star \psi \in C^\infty(\mathbb{R}^d)$ with

$$D^\alpha(u \star \psi) = u \star (D^\alpha \psi)$$

For the second equality, observe that

$$D^\alpha(\tau_x \check{\psi}) = D^\alpha(\psi(x - \cdot)) = (-1)^{|\alpha|}(D^\alpha \psi)(x - \cdot) = (-1)^{|\alpha|}\tau_x (D^\alpha \psi)^\vee$$

and thus

$$(u \star (D^\alpha \psi))(x) = u[\tau_x (D^\alpha \psi)^\vee] = (-1)^{|\alpha|} u[D^\alpha(\tau_x \check{\psi})] = (D^\alpha u)[\tau_x \check{\psi}] = ((D^\alpha u) \star \psi)(x0$$

for all $x \in \mathbb{R}^d$.

(ii) Suppose $x \in \mathbb{R}^d$ satisfies $(u \star \psi)(x) \neq 0$. By Lemma 5.8, we must have $\operatorname{supp}(u) \cap \operatorname{supp}(\tau_x \check{\psi}) \neq \varnothing$. In particular, we can pick $z \in \operatorname{supp} u$ such that $(\tau_x \check{\psi})(z) = \psi(x - z) \neq 0$ and thus $x - z \in \operatorname{supp} \psi$. It follows that

$$\operatorname{supp}(u \star \psi) = \overline{\{x \in \mathbb{R}^d \colon (u \star \psi)(x) \neq 0\}} \subset \operatorname{supp} u + \operatorname{supp} \psi$$

where we have also used the fact that $\operatorname{supp} u + \operatorname{supp} \psi$ is closed. ■

Definition Convolution of distributions
Let $u, v \in \mathcal{D}'(\mathbb{R}^d)$ with v compactly supported. We define their convolution $u \star v \in \mathcal{D}'(\mathbb{R}^d)$ to be the unique distribution such that

$$(u \star v) \star \varphi = u \star (v \star \varphi) \qquad \forall \varphi \in \mathcal{D}(\mathbb{R}^d)$$

Lemma 5.11
Let $f \in L^1_{\text{loc}}(\mathbb{R}^d)$ and $g \in L^1(\mathbb{R}^d)$, with g compactly supported. Then

$$T_{f\star g} = T_f \star T_g$$

Proof. For $\varphi \in \mathcal{D}(\mathbb{R}^d)$, we have

$$\begin{aligned}((T_f \star T_g) \star \varphi)(x) &= (T_f \star (T_g \star \varphi))(x) \\ &= \int f(t)(T_g \star \varphi)(x+t)\,dt \\ &= \iint f(t)g(s)\varphi(x+t+s)\,ds\,dt \\ &= \iint f(t)g(s-t)\varphi(x+s)\,ds\,dt \\ &= \int (f\star g)(s)\varphi(x+s)\,ds \\ &= (T_{f\star g} \star \varphi)(x)\end{aligned}$$

for all $x \in \mathbb{R}^d$. Above, we used Fubini to swap the order of integration. To justify this, pick $R > 0$ sufficiently large such that $\operatorname{supp} g \subset B(0,R)$ and $\operatorname{supp}\varphi \subset B(x,R)$. We then estimate

$$\begin{aligned}\iint |f(t)g(s)\varphi(x+t+s)|\,ds\,dt &= \int_{B(0,R)\times B(0,2R)} |f(t)g(s)\varphi(x+t+s)|\,ds\,dt \\ &\leq ||g||_1||\varphi||_\infty \int_{B(0,2R)} |f(t)|\,dt\end{aligned}$$

with $\int_{B(0,2R)} |f(t)|\,dt < \infty$ since $f \in L^1_{\text{loc}}(\mathbb{R}^d)$. ■

Lemma 5.12
Let $u, v \in \mathcal{D}'(\mathbb{R}^d)$ with v compactly supported. Then

$$D^\alpha(u\star v) = (D^\alpha u)\star v = u \star (D^\alpha v)$$

for any $\alpha \in \mathbb{Z}^d_{\geq 0}$.

Proof. For any $\varphi \in \mathcal{D}(\mathbb{R}^d)$, we have

$$D^\alpha(u\star v)\star\varphi = (u\star v)\star D^\alpha\varphi = u\star(v\star D^\alpha\varphi) = u\star(D^\alpha v\star\varphi) = (u\star D^\alpha v)\star\varphi$$

Similarly, we have

$$(u\star D^\alpha v)\star\varphi = u\star(D^\alpha v\star\varphi) = u\star D^\alpha(v\star\varphi) = (D^\alpha u)\star(v\star\varphi)$$

as required. ■

We now apply the theory we have established to the analysis of partial differential equations.

Definition Fundamental solution
Let $k \in \mathbb{N}$ and $a_\alpha \in \mathbb{C}$ for each $\alpha \in \mathbb{Z}^d_{\geq 0}$ with $|\alpha| \leq k$. We say that $G \in \mathcal{D}'(\mathbb{R}^d)$ is a fundamental solution of the partial differential operator

$$L := \sum_{|\alpha|\leq k} a_\alpha D^\alpha$$

if it satisfies the distributional equation

$$LG = \delta_0$$

Theorem 5.13
If G is a fundamental solution of L and $u_0 \in \mathcal{D}'(\mathbb{R}^d)$ has compact support, then $u := G \star u_0$ is a distributional solution to the PDE $Lu = u_0$.

Proof. Observe that

$$L(G \star u_0) = \sum_{|\alpha| \leq k} a_\alpha D^\alpha (G \star u_0) = \sum_{|\alpha| \leq k} a_\alpha (D^\alpha G) \star u_0 = LG \star u_0 = \delta_0 \star u_0 = u_0$$

as required. ■

Example *Poisson equation*
Suppose we want to solve the PDE

$$\Delta u = f \qquad \text{on } \mathbb{R}^3$$

with $f \in L^1(\mathbb{R}^3)$ having compact support. To do this, consider the function

$$\begin{aligned} g \colon \mathbb{R}^3 &\to \mathbb{R} \\ x &\mapsto \begin{cases} \dfrac{1}{4\pi|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases} \end{aligned}$$

Note that $g \in L^1_{\text{loc}}(\mathbb{R}^3)$, so $G := T_g \in \mathcal{D}'(\mathbb{R}^3)$. You will show on Example Sheet 4 that $\Delta G = \delta_0$ and thus $u := G \star T_f$ is a *distributional solution* of the Poisson equation. If it turns out that $u \in C^2(\mathbb{R}^3)$, then we say that it is a *classical solution.*

5.4 Tempered distributions and the Fourier transform

Now that we have extended many operations like differentiation and convolution to distributions, it is also natural to try extending the Fourier transform. Unfortunately, working with $\mathcal{D}$ and $\mathcal{D}'$ is not ideal since $\mathcal{D}$ is not closed under a Fourier transform — the Fourier transform of a compactly supported function is, in fact, never compactly supported. It turns out that the right space of 'test functions' in this case is the Schwartz space $\mathcal{S}$, which admit a natural extension of the Fourier transform for so-called *tempered distributions.*

On $\mathcal{S}(\mathbb{R}^d)$, we can define a family of norms $\{p_N \colon N \in \mathbb{N}\}$ by

$$p_N(f) := \max_{|\alpha| \leq N} \sup_{x \in \mathbb{R}^d} \left|(1 + |x|)^N (D^\alpha f)(x)\right|$$

This family of norms induces a topology on $\mathcal{S}(\mathbb{R}^d)$, which is metrizable via

$$\operatorname{dist}(f, g) = \sum_{N=1}^{\infty} \min(p_N(f - g), 2^{-N})$$

Equivalently, the same topology is generated the family of seminorms $\{\rho_{N,\alpha} \colon N \in \mathbb{N}, \alpha \in \mathbb{Z}^d_{\geq 0}\}$ given by

$$\rho_{N,\alpha}(f) = \sup_{x \in \mathbb{R}^d} \left|(1 + |x|)^N (D^\alpha f)(x)\right|$$

Definition Space of tempered distributions
The space $\mathcal{S}'(\mathbb{R}^d)$ is defined as the space of continuous linear functionals on $\mathcal{S}(\mathbb{R}^d)$. We call its elements tempered distributions.

Remark. As a dual space, $\mathcal{S}'(\mathbb{R}^d)$ is naturally endowed with the weak-$*$ topology.

Lemma 5.14 Continuity of a linear functional
Let $u\colon \mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$ be a linear functional. The following are equivalent:
(i) $u \in \mathcal{S}'(\mathbb{R}^d)$
(ii) $u[\varphi_j] \to u[\varphi]$ *in* $\mathbb{C}$ *whenever* $\varphi_j \to \varphi$ *in* $\mathcal{S}(\mathbb{R}^d)$.
(iii) *There exist constants* $C > 0$ *and* $N \in \mathbb{N}$ *such that* $|u[\varphi]| \leq Cp_N(\varphi)$ *for all* $\varphi \in \mathcal{S}(\mathbb{R}^d)$.

Proof. (i) $\implies$ (ii): Immediate from metrizability of topology.
(ii) $\implies$ (iii): Suppose, on the contrary, that no such C, N exist. Then, for each $n \in \mathbb{N}$, we can pick $\varphi_n \in \mathcal{S}(\mathbb{R}^d)$ such that $|u[\varphi_n]| > np_n(\varphi_n)$. For any fixed $M \in \mathbb{N}$, we have that, for each $n \geq M$,

$$p_M\left(\frac{\varphi_n}{|u[\varphi_n]|}\right) \leq p_n\left(\frac{\varphi_n}{|u[\varphi_n]|}\right) < \frac{1}{n} \to 0$$

Thus, $\varphi_n/|u[\varphi_n]| \to 0$ in $\mathcal{S}(\mathbb{R}^d)$. By hypothesis, $u[\frac{\varphi_n}{|u[\varphi_n]|}] \to 0$. However, this contradicts the fact that $|u[\frac{\varphi_n}{|u[\varphi_n]|}]| = 1$ for all $n \in \mathbb{N}$.
(iii) $\implies$ (i): For any $\varepsilon > 0$, we can set $\delta = \varepsilon/C$. Then we have that $|p_N(\varphi_1 - \varphi_2)| < \delta \implies |u[\varphi_1 - \varphi_2]| < \varepsilon$ and thus u is continuous. ■

Lemma 5.15 Functions as tempered distributions
Let $f\colon \mathbb{R}^d \to \mathbb{C}$ be a measurable function such that there exists $C, N > 0$ such that $|f(x)| \leq C(1 + |x|)^N$ for a.e. $x \in \mathbb{R}^d$. Then

$$\begin{aligned} T_f\colon \mathcal{S} &\to \mathbb{C} \\ g &\mapsto \int fg\,dx \end{aligned}$$

is a well-defined tempered distribution.

Proof. Suppose $g_n \to g$ in $\mathcal{S}$. Then $p_{N+d+1}(g_n - g) \to 0$, so we have

$$\begin{aligned} \int |f \cdot (g_n - g)|\,dx &\leq \int C(1 + |x|)^{N+d+1}|g_n(x) - g(x)|(1 + |x|)^{-d-1}\,dx \\ &\leq Cp_{N+d+1}(g_n - g)\int (1 + |x|)^{-d-1}\,dx \to 0 \end{aligned}$$

as required. ■

Definition Translate of a tempered distribution
Let $t \in \mathbb{R}^d$. We define $\tau_t u \in \mathcal{S}'(\mathbb{R}^d)$ by

$$(\tau_t u)[\varphi] := u[\tau_{-t}\varphi] \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^d)$$

Lemma 5.16
Let f be as in Lemma 5.15. For each $t \in \mathbb{R}^d$, we have

$$\tau_t T_f = T_{\tau_t f}$$

Proof. Note that

$$(\tau_t T_f)[\varphi] = T_f[\tau_{-t}\varphi] = \int f(x)\varphi(x + t)\,dx = \int f(x - t)\varphi(t)\,dx = T_{\tau_t f}[\varphi]$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^d)$. ■

Definition Reflection of a tempered distribution
For $u \in \mathcal{S}'(\mathbb{R}^d)$, we define $\check{u} \in \mathcal{S}'(\mathbb{R}^d)$ by

$$\check{u}[\varphi] = u[\check{\varphi}] \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^d)$$

Lemma 5.17
Let f be as in Lemma 5.15. Then

$$(T_f)^\vee = T_{\check{f}}$$

Proof. Observe that

$$(T_f)^\vee[\varphi] = T_f[\check{\varphi}] = \int f(x)\varphi(-x)\,dx = \int f(-x)\varphi(x)\,dx = T_{\check{f}}[\varphi]$$

for every $\varphi \in \mathcal{S}(\mathbb{R}^d)$. ■

Definition Fourier transform
Let $u \in \mathcal{S}'(\mathbb{R}^d)$. We define the Fourier transform of u to be

$$\widehat{u}[\varphi] := u[\widehat{\varphi}] \qquad \forall\, \varphi \in \mathcal{S}(\mathbb{R}^d)$$

Proposition 5.18
(i) *The map $\varphi \mapsto \widehat{\varphi}$ is continuous on $\mathcal{S}(\mathbb{R}^d)$.*
(ii) *The map $u \mapsto \widehat{u}$ is continuous on $\mathcal{S}'(\mathbb{R}^d)$.*

Proof. For (i), suppose $\varphi_n \to \varphi$ in $\mathcal{S}(\mathbb{R}^d)$. Then, for all $N \in \mathbb{N}$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$,

$$\rho_{N,\alpha}(f_n - f) = \sup_{x\in\mathbb{R}^d} |(1+|x|)^N (D^\alpha(f_n - f))(x)| \to 0$$

Fix $k \in \mathbb{N}$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$. By Proposition 4.1, we have

$$\sup_{\xi\in\mathbb{R}^d} \left| |\xi|^{2k} D^\alpha(\widehat{f_n} - \widehat{f}) \right| \leq (2\pi)^{|\alpha|-2k} \int_{\mathbb{R}^d} |\Delta^k(x^\alpha(f_n - f))|\,dx \tag{†}$$

Observe that

$$\begin{aligned}\int_{\mathbb{R}^d} |x^\gamma D^\beta(f_n - f)|\,dx &\leq \int_{\mathbb{R}^d} (1+|x|)^{|\eta|} |D^\beta(f_n - f)|\,dx \\ &\leq \sup_{x\in\mathbb{R}^d} \left|(1+|x|)^{d+1+|\eta|} D^\beta(f_n - f)\right| \cdot \int_{\mathbb{R}^d} (1+|x|)^{-(d+1)}\,dx \\ &\to 0\end{aligned}$$

noting that the integral in the penultimate line is finite. Since the RHS in (†) is simply a linear combination of integrals of the form $\int_{\mathbb{R}^d} |x^\gamma D^\beta(f_n - f)|\,dx$, it follows that

$$\sup_{\xi\in\mathbb{R}^d} \left| |\xi|^{2k} D^\alpha(\widehat{f_n} - \widehat{f}) \right| \to 0$$

and thus $\widehat{f_n} \to \widehat{f}$ in $\mathcal{S}(\mathbb{R}^d)$.

For (ii), suppose $u_n \overset{*}{\rightharpoonup} u$. For any $\varphi \in \mathcal{S}(\mathbb{R}^d)$, we have

$$\widehat{u_n}(\varphi) = u_n(\widehat{\varphi}) \to u(\widehat{\varphi}) = \widehat{u}(\varphi)$$

and thus $\widehat{u_n} \overset{*}{\rightharpoonup} \widehat{u}$. ■

Remark. Strictly speaking, we have only proven sequential continuity in (ii).

Lemma 5.19
Let $f \in L^1(\mathbb{R}^d) \cup L^2(\mathbb{R}^d)$. Then

$$\widehat{T_f} = T_{\widehat{f}}$$

Proof. If $f \in L^1(\mathbb{R}^d)$, then

$$\widehat{T_f}(\varphi) = \int f\widehat{\varphi}\,dx = \iint f(x)\varphi(y)e^{-2\pi i\langle y,x\rangle}\,dy\,dx = \int \widehat{f}(y)\varphi(y)\,dy$$

for all $\varphi \in \mathcal{S}(\mathbb{R}^d)$. Above, we used Fubini's theorem, which is justified by the fact that $f, h \in L^1(\mathbb{R}^d)$.

If $f \in L^2(\mathbb{R}^d)$, then the identity can be proven through the density of $\mathcal{S}(\mathbb{R}^d)$ in $L^2(\mathbb{R}^d)$. Alternatively, one can also use Parseval's formula. ■

Lemma 5.20 Basic properties of Fourier transform
Let $u \in \mathcal{S}'(\mathbb{R}^d)$, $t \in \mathbb{R}^d$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$. Then

(i) $(e^{2\pi i\langle x,t\rangle}u)^\vee = \tau_x\widehat{u}$
(ii) $\widehat{\tau_t u} = e^{-2\pi i\langle \xi,t\rangle}\widehat{u}$
(iii) $\widehat{D^\alpha u} = (2\pi i)^{|\alpha|}\xi^\alpha\widehat{u}$
(iv) $D^\alpha\widehat{u} = (-2\pi i)^{|\alpha|}\widehat{x^\alpha u}$
(v) $\widehat{\widehat{u}} = \check{u}$

Exercise. Prove the above lemma.

5.5 Periodic distributions

Definition Space of periodic test functions
We define the space $\mathcal{D}(\mathbb{T}^d)$ by

$$\mathcal{D}(\mathbb{T}^d) := \left\{ f \in C^\infty(\mathbb{R}^d) \colon \tau_a f = f \ \forall\, a \in \mathbb{Z}^d \right\}$$

endowed with the metric

$$\operatorname*{dist}_{\mathcal{D}}(f,g) := \sum_\alpha \min\left(||D^\alpha(f-g)||_\infty, 2^{-|\alpha|} \right)$$

Definition Space of periodic distributions
We define $\mathcal{D}'(\mathbb{T}^d)$ as the space of continuous linear functionals on $\mathcal{D}(\mathbb{T}^d)$.

Lemma 5.21 Periodic functions as periodic distributions
For $f \in L^1(\mathbb{T}^d)$,

$$\begin{aligned} T_f \colon \mathcal{D}(\mathbb{T}^d) &\to \mathbb{C} \\ \varphi &\mapsto \int_{[0,1]^d} f(x)\varphi(x)\,dx \end{aligned}$$

defines an element of $\mathcal{D}'(\mathbb{T}^d)$.

Definition Fourier coefficient
Let $u \in \mathcal{D}'(\mathbb{T}^d)$. We define its Fourier coefficients as

$$\widehat{u}(\xi) = u[e^{-2\pi i\langle \cdot,\xi\rangle}], \qquad \xi \in \mathbb{Z}^d$$

Lemma 5.22
Let $u \in \mathcal{D}'(\mathbb{T}^d)$. Then there exists $N \in \mathbb{Z}_{\geq 0}$ such that

$$|\widehat{u}(\xi)| \leq (2+|\xi|)^N \qquad \forall\, \xi \in \mathbb{Z}^d$$

for all $\xi \in \mathbb{Z}_{\geq 0}$.

Proof. Suppose, on the contrary, that, for every $N \in \mathbb{Z}_{\geq 0}$, there exists a sequence (ξ_n) in $\mathbb{Z}^d$ such that $|\widehat{u}(\xi_n)| > (2 + |\xi_n|)^n$ for all n. Observe that

$$||D^\alpha e^{-2\pi i\langle \cdot, \xi\rangle}||_\infty = ||(-2\pi i)^{|\alpha|}\xi^\alpha e^{-2\pi i\langle \cdot, \xi\rangle}||_\infty \leq (2\pi|\xi|)^{|\alpha|}$$

It follows that

$$f_n := (2 + |\xi_n|)^{-n} e^{-2\pi i\langle \cdot, \xi_n\rangle} \to 0$$

in $\mathcal{D}(\mathbb{T}^d)$. On the other hand, we have $|u[f_n]| > 1$ for all $n \in \mathbb{N}$, contradicting the fact that $u[f_n] \to 0$ by continuity of u. ■

Proposition 5.23 Convergence of Fourier series
For every $u \in \mathcal{D}'(\mathbb{T}^d)$, we have

$$\sum_{|\xi| \leq k} \widehat{u}(\xi) T_{e^{2\pi i\langle \cdot, \xi\rangle}} \overset{*}{\rightharpoonup} u$$

and we then write

$$u = \sum_{\xi \in \mathbb{Z}^d} \widehat{u}(\xi) e^{2\pi i\langle \cdot, \xi\rangle}$$

Proof. Fix $\varphi \in \mathcal{D}(\mathbb{T}^d)$. Observe that

$$\begin{aligned}
\sum_{|\xi| \leq k} \widehat{u}(\xi) T_{e^{2\pi i\langle \cdot, \xi\rangle}}[\varphi] &= \sum_{|\xi| \leq k} \widehat{u}(\xi) \int \varphi(x) e^{2\pi i\langle x, \xi\rangle}\, dx \\
&= \sum_{|\xi| \leq k} u[e^{-2\pi i\langle \cdot, \xi\rangle}]\widehat{\varphi}(-\xi) \\
&= u\left[\sum_{|\xi| \leq k} \widehat{\varphi}(-\xi) e^{2\pi i\langle \cdot, -\xi\rangle}\right] \\
&\to u[\varphi]
\end{aligned}$$

using the definition of Fourier transform, linearity of u, Fourier inversion (Theorem 4.9) and continuity of u. ■

Now, recall the linear map

$$\begin{aligned}
P\colon L^1(\mathbb{R}^d) &\to L^1(\mathbb{T}^d) \\
f &\mapsto \sum_{a \in \mathbb{Z}^d} \tau_a f
\end{aligned}$$

Restricting to $\mathcal{S}(\mathbb{R}^d)$, we get a continuous map

$$P\colon \mathcal{S}(\mathbb{R}^d) \to \mathcal{D}(\mathbb{T}^d)$$

It turns out that $P\colon \mathcal{S}(\mathbb{R}^d) \to \mathcal{D}(\mathbb{T}^d)$ is surjective. To show this, consider a function $f \in C_c^\infty(\mathbb{R}^d)$ such that $f \geq 0$ everywhere and $f > 0$ on $[0, 1]^d$. Then $Pf > 0$ everywhere. Then $\chi(x) := f(x)/Pf(x)$ defines a $C_c^\infty(\mathbb{R}^d)$ function such that $P\chi = 1$. It then follows that, for any $g \in \mathcal{D}(\mathbb{T}^d)$, we have $\chi g \in \mathcal{S}(\mathbb{R}^d)$ with $P[\chi g] = g$.

We define the adjoint map $P^*\colon \mathcal{D}'(\mathbb{T}^d) \to \mathcal{S}'(\mathbb{R}^d)$ by

$$(P^* u)[f] := u[Pf] \qquad \forall\, f \in \mathcal{D}(\mathbb{T}^d)$$

Lemma 5.24
P^* is linear, injective and continuous with respect to the weak-$*$ topologies.

Proof. P^* is linear since it is a composition of linear maps. For any $u \in \mathcal{D}'(\mathbb{T}^d)\backslash\{0\}$, we can pick $f \in \mathcal{D}(\mathbb{T}^d)$ such that $u[f] \neq 0$ and thus

$$(P^*u)[\chi f] = u[P\chi f] = u(f) \neq 0$$

so P^* is injective. Finally, suppose $u_n \overset{*}{\rightharpoonup} u$ in $\mathcal{D}'(\mathbb{T}^d)$. For any $f \in \mathcal{S}(\mathbb{R}^d)$, we have

$$(P^*u_n)[f] = u_n[Pf] \to u[Pf] = (P^*u)[f]$$

so $P^*u_n \overset{*}{\rightharpoonup} P^*u$. ■

Corollary 5.25
Suppose $v \in P^*(\mathcal{D}'(\mathbb{T}^d)) \subset \mathcal{S}'(\mathbb{R}^d)$. Then

$$v(x) = \sum_{\xi \in \mathbb{Z}^d} v[\chi \cdot e^{-2\pi i\langle \cdot, \xi\rangle}]e^{2\pi i\langle x,\xi\rangle}$$

Proof. First, observe that, for any $u \in \mathcal{D}'(\mathbb{T}^d)$,

$$(P^*u)[\chi e^{-2\pi i\langle \cdot,\xi\rangle}] = u[P\chi e^{-2\pi i\langle \cdot,\xi\rangle}] = u[e^{-2\pi i\langle \cdot,\xi\rangle}] = \widehat{u}(\xi)$$

for all $\xi \in \mathbb{Z}^d$. Moreover, for any $\varphi \in \mathcal{S}(\mathbb{R}^d)$, we have

$$(P^*T_{e^{2\pi i\langle \cdot,\xi\rangle}})[\varphi] = \int_{[0,1]^d} e^{2\pi i\langle x,\xi\rangle} P[\varphi](x)\,dx = \int_{\mathbb{R}^d} e^{2\pi i\langle x,\xi\rangle}\varphi(x)\,dx = T_{e^{2\pi i\langle \cdot,\xi\rangle}}$$

for all $\xi \in \mathbb{Z}^d$. ■

Definition
Let $N \in \mathbb{Z}_{\geq 0}$. For each sequence (x_n) in $\mathbb{R}^d$ and each sequence (b_j) in $\mathbb{C}$ satisfying

$$\sum_{\substack{j\in\mathbb{N}\\ |x_j|<R}} |b_j| \leq R^N$$

for all $R \in \mathbb{N}$, we define $\sum_j a_j\delta_j \in \mathcal{S}'(\mathbb{R}^d)$ by

$$\left(\sum_j b_j\delta_j\right)[f] = \sum_j b_j f(x_j) \qquad \forall\, f \in \mathcal{S}(\mathbb{R}^d)$$

Remark. Note that $\sum_j b_j\delta_j$ may not be measure.

Corollary 5.26
If $v \in P^*(\mathcal{D}'(\mathbb{T}^d)) \subset \mathcal{S}'(\mathbb{R}^d)$, then

$$\widehat{v} = \sum_{\xi\in\mathbb{Z}^d} v[\chi e^{-2\pi i\langle \cdot,\xi\rangle}]\delta_\xi$$

Proof. It is immediate from definitions that

$$\sum_{|\xi|\leq N} v[\chi e^{-2\pi i\langle x,\xi\rangle}]e^{2\pi i\langle x,\xi\rangle} \overset{*}{\rightharpoonup} \sum_{\xi\in\mathbb{Z}^d} v[\chi e^{-2\pi i\langle x,\xi\rangle}]e^{2\pi i\langle x,\xi\rangle}$$

as $N \to \infty$. By continuity of the Fourier transform on $\mathcal{S}'(\mathbb{R}^d)$, it follows that

$$\widehat{T_{e^{2\pi i\langle \cdot,\xi\rangle}}}[\varphi] = \int_{\mathbb{R}^d} e^{2\pi i\langle x,\xi\rangle}\widehat{\varphi}(x)\,dx = \varphi(\xi) = \delta_\xi[\varphi]$$

as required. ■

Definition Periodic tempered distribution
We say that $u \in \mathcal{S}'(\mathbb{R}^d)$ is periodic if $\tau_a v = v$ for all $a \in \mathbb{Z}^d$.

Proposition 5.27 Equivalent definition
Let $v \in \mathcal{S}'(\mathbb{R}^d)$. Then $v \in P^*(\mathcal{D}'(\mathbb{T}^d))$ if and only if v is periodic.

Proof. ($\Longrightarrow$) Let $a \in \mathbb{Z}^d$. For each $u \in \mathcal{D}'(\mathbb{T}^d)$, we have

$$(\tau_a(P^*u))[\varphi] = u[\tau_{-a}P\varphi] = u[P\varphi] = (P^*u)[\varphi]$$

for all $\varphi \in \mathcal{S}(\mathbb{R}^d)$.

($\Longleftarrow$) Fix a periodic tempered distribution $v \in \mathcal{S}'(\mathbb{R}^d)$. Define $u \in \mathcal{D}'(\mathbb{T}^d)$ by $u[\varphi] := v[\chi\varphi]$ for all $\varphi \in \mathcal{D}(\mathbb{T}^d)$. Observe that

$$(P * u)[\psi] = u[P[\psi]] = u[\chi P[\psi]]$$

for all $\psi \in \mathcal{S}(\mathbb{R}^d)$. Since $P[\chi] \equiv 1$, we can write

$$\psi(x) = \sum_{a\in\mathbb{Z}^d} (\tau_{-a}\chi)(x)\psi(x) \qquad \forall\, x \in \mathbb{R}^d$$

Since v is periodic, we have that

$$v[(\tau_{-a}\chi)f] = v[\tau_a((\tau_{-a}\chi)f)] = v[\chi\tau_a f]$$

Moreover, we have from the definition of P that

$$(\chi P[\psi])(x) = \sum_{a\in\mathbb{Z}^d} \chi(x)(\tau_a\psi)(x) \qquad \forall\, x \in \mathbb{R}^d$$

It remains to show that the two series above converge in $\mathcal{S}(\mathbb{R}^d)$. We will only do this for the first one; a similar argument works for the second. The derivative $D^\alpha((\tau_{-a}\chi)\psi)$ is a linear combination of products of a derivative of $\tau_{-a}\chi$ and a derivative of ψ. The order of the derivatives are at most $|\alpha|$. The derivatives of $\tau_{-a}\chi$ are bounded by a constant depending only on $|\alpha|$ and χ. Also, $(\tau_{-a}\chi)(x)\psi(x) = 0$ if $|x| \le |a| - C_1$ for a constant $C_1 = C_1(\chi)$. This implies that $|x| + 1 \ge |a|/C_1$ on the support of $(\tau_{-a}\chi)\cdot(f)$. It then follows that

$$p_N((\tau_{-a}\chi)\cdot f) \le C_2(N)p_{N+d+1}(f)\frac{C_1^{d+1}}{|a|^{d+1}}$$

for some constant $C_2(N)$ that comes from the contribution of the derivatives of $\tau_{-a}\chi$. ■

Example *Poisson summation formula for tempered distributions*
Let $v = \sum_{a\in\mathbb{Z}^d} \delta_a \in \mathcal{S}'(\mathbb{R}^d)$. Note that

$$v[\chi \cdot e^{-2\pi i\langle\cdot,\xi\rangle}] = \sum_{a\in\mathbb{Z}^d} \chi(a)e^{-2\pi i\langle a,\xi\rangle} = (P[\chi])(0) = 1$$

for all $\xi \in \mathbb{R}^d$. By Corollary 5.26, we conclude that

$$\widehat{u} = u$$

which is the Poisson summation formula for tempered distributions.

6 Sobolev spaces and PDEs

6.1 Basic definitions of Sobolev spaces

Let $\Omega \subset \mathbb{R}^d$ be an open set.

Definition Weak derivative
Let $f \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{Z}^d_{\geq 0}$. We say that f has a weak derivative of order α if there exists $g \in L^1_{\text{loc}}(\Omega)$ such that

$$D^\alpha T_f = T_g.$$

Remark. It can be shown that a weak derivative $g \in L^1_{\text{loc}}(\Omega)$ of $f \in L^1_{\text{loc}}(\Omega)$ is unique if it exists.

Notation. We will often write $D^\alpha f$ for g.

Definition Sobolev space
Let $p \in [1, \infty]$ and $k \in \mathbb{N}$. The Sobolev space $W^{k,p}(\Omega)$ is defined as

$$W^{k,p}(\Omega) := \{f \in L^p(\Omega) \colon D^\alpha f \in L^p(\Omega)\ \forall\, |\alpha| \leq k\}$$

endowed with the norm

$$||f||_{W^{k,p}(\Omega)} := \sum_{|\alpha| \leq k} ||D^\alpha f||_p$$

Remark. Roughly speaking, it is helpful intuition to think of Sobolev spaces with larger k as containing smoother functions while those with larger p contain less spiky functions.

In the case $p = 2$, we often use the notation

$$H^k(\Omega) := W^{k,2}(\Omega)$$

In the literature, these are sometimes called *L^2-based Sobolev spaces.*

In the case $\Omega = \mathbb{R}^d$, Sobolev spaces have an alternative characterisation via the Fourier transform.

Proposition 6.1
Let $f \in L^2(\mathbb{R}^d)$ and $k \in \mathbb{Z}_{\geq 0}$. Then $f \in H^k(\mathbb{R}^d)$ iff

$$||f||_{H^k} := \left(\int |\widehat{f}(\xi)|^2 (1 + |\xi|^2)\, d\xi \right)^{1/2} < \infty$$

Moreover, the norms $||\cdot||_{W^{k,2}}$ and $||\cdot||_{H^k}$ are equivalent.

To prove this proposition, we first need the following lemma from Fourier analysis.

Lemma 6.2
Let $f \in L^2(\mathbb{R}^d)$. If f has a weak derivative $D^\alpha f \in L^2(\mathbb{R}^d)$, then

$$\widehat{D^\alpha f}(\xi) = (2\pi i)^{|\alpha|} \xi^\alpha \widehat{f}(\xi)$$

Conversely, if the map $\xi \mapsto \xi^\alpha \widehat{f}(\xi)$ is in $L^2(\mathbb{R}^d)$, then f has a weak derivative $D^\alpha f \in L^2(\mathbb{R}^d)$.

Exercise. Prove the above lemma.

Proof of Proposition 6.1. ($\Longrightarrow$) Suppose $f \in W^{k,2}(\mathbb{R}^d)$. Then for any multi-index $\alpha \in \mathbb{Z}^d_{\geq 0}$ with $|\alpha| \leq k$, we have

$$\int |2\pi i|^{2|\alpha|} |\xi|^\alpha |\widehat{f}(\xi)|^2\, d\xi = ||D^\alpha f||^2_{L^2} \leq ||f||^2_{W^{k,2}}$$

Since the $||f||_{H^k}^2$ is simply a linear combination of terms of the form $||D^\alpha f||_{L^2}^2$ for $|\alpha| \leq k$ (with coefficients independent of f), it follows that there exists $C = C(d,k) > 0$ such that $||f||_{H^k} \leq C||f||_{W^{k,2}}$ for all $f \in W^{k,2}(\mathbb{R}^d)$.

($\Longleftarrow$) Suppose $||f||_{H^k} < \infty$. For any $\alpha \in \mathbb{Z}_{\geq 0}^d$ with $|\alpha| \leq k$¡ we have

$$||D^\alpha f||_{L^2} = (2\pi)^{|\alpha|}||\xi^\alpha \widehat{f}(\xi)||_{L^2} \leq (2\pi)^{|\alpha|}||f||_{H^k}$$

It then follows that there exists a constant $\tilde{C} = \tilde{C}(d,k)$ such that $||f||_{W^{k,p}} \leq \tilde{C}||f||_{H^k}$ for all $f \in L^2$ with $||f||_{H^k} < \infty$. ■

Of course, the norm $||\cdot||_{H^k}$ makes sense even if k is not an integer. This motivates the following definition:

Definition H^s spaces
Let $s \in \mathbb{R}$. The Sobolev space $H^s(\mathbb{R}^d)$ are defined by

$$H^s(\mathbb{R}^d) := \left\{ u \in \mathcal{S}'(\mathbb{R}^d) \colon \widehat{u} \in L^2_{\text{loc}}(\mathbb{R}^d), ||u||_{H^s(\mathbb{R}^d)} < \infty \right\}$$

where the norm is given by

$$||u||_{H^s(\mathbb{R}^d)} := \int |\widehat{u}(\xi)|^2 (1+|\xi|^2)^s \, d\xi$$

A particularly nice property of H^s spaces is that they are Hilbert spaces equipped with the inner product

$$\langle f, g \rangle_{H^s} := \int (1+|\xi|^2)^s \widehat{f}(\xi)\overline{\widehat{g}(\xi)} \, d\xi$$

In the case $s = 1$, we have the usual formula

$$\langle f, g \rangle_{H^1} := \int f(x)\overline{g(x)} + \frac{1}{4\pi^2}(\nabla f) \cdot \overline{(\nabla g)} \, dx$$

Notational remark. In the functional-analytic literature, one may sometimes find that the $||\cdot||_{H^s}$ norm is written as

$$||u||_{H^s(\mathbb{R}^d)} := \int |\widehat{u}(\xi)|^2 \langle \xi \rangle^{2s} \, d\xi$$

where $\langle \xi \rangle := (1+|\xi|^2)^{1/2}$ is known as the *Japanese bracket.*

Proposition 6.3 Mollification
Let $\varphi \in C_c^\infty(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d} \varphi(x) \, dx = 1$ and $\varphi \geq 0$. For each $n \in \mathbb{N}$, define

$$\varphi_n(x) = n^d \varphi(nx)$$

Then $\varphi_r \star f \to f$ in $H^s(\mathbb{R}^d)$ for every $f \in H^s(\mathbb{R}^n)$.

Proof. Fix $f \in H^s(\mathbb{R}^n)$. Using Proposition 4.1, we have that

$$\widehat{\varphi_n \star f}(\xi) = \widehat{\varphi}(n^{-1}\xi) \cdot \widehat{f}(\xi)$$

Since $|\widehat{\varphi}(n^{-1}\xi) \cdot \widehat{f}(\xi)| \leq |\widehat{f}(\xi)|$ and $\widehat{\varphi_n \star f}(\xi) \to \widehat{f}(\xi)$ for each ξ, the claim follows by dominated convergence. ■

6.2 Sobolev embedding

In PDE analysis, we often obtain weak solutions in some Sobolev space that may not be sufficiently smooth to be a classical solution to the PDE. Nevertheless, as we will show in this subsection, it turns out that, if s is sufficiently large, H^s embeds into C^k.

Definition Hölder space
Let $k \in \mathbb{Z}_{\geq 0}$ and $\gamma \in [0,1]$. The Hölder space $C^{k,\gamma}(\mathbb{R}^d)$ is defined as

$$C^{k,\gamma}(\mathbb{R}^d) := \left\{ f \in C^k(\mathbb{R}^d) \colon ||f||_{C^{k,\gamma}(\mathbb{R}^d)} < \infty \right\}$$

where the norm is defined by

$$||f||_{C^{k,\gamma}(\mathbb{R}^d)} := \sum_{|\alpha| \leq k} ||D^\alpha f||_\infty + \sum_{|\alpha|=k} \sup_{x,y \in \mathbb{R}^d} \frac{|(D^\alpha f)(x) - (D^\alpha f)(y)|}{|x-y|^\gamma}$$

Definition Continuous embedding
Let $(X, ||\cdot||_X)$ and $(Y, ||\cdot||_Y)$ be Banach spaces. We say that X continuously embeds into Y if $X \subset Y$ and there exists $C > 0$ such that $||f||_X \leq C||f||_Y$ for all $f \in X$.

Theorem 6.4 Sobolev embedding theorem

(i) *Let $1 \leq p \leq q < \infty$ and $k, l \in \mathbb{Z}_{\geq 0}$ satisfy $k \geq l$. If*

$$\frac{l}{d} \geq \frac{1}{p} - \frac{1}{q},$$

then $W^{k,p}(\mathbb{R}^d)$ embeds continuously into $W^{k-l,q}(\mathbb{R}^d)$.

(ii) *Let $p \in [1, \infty)$, $k, l \in \mathbb{Z}_{\geq 0}$ with $k \geq l$ and $\gamma \in (0,1)$ such that*

$$\frac{l-\gamma}{d} \geq \frac{1}{p}.$$

Then $W^{k,p}(\mathbb{R}^d)$ embeds continuously into $C^{k-l,\gamma}(\mathbb{R}^d)$.

Remarks

- In (ii), the conclusion means that there is an element in the equivalence class that belongs to $C^{k-l,\gamma}(\mathbb{R}^d)$.
- The above theorem remains true when replacing $\mathbb{R}^d$ with a 'sufficiently nice' domain Ω. However, these versions of the theorem are beyond the scope of this course.

In this course, we will only be proving the Sobolev embedding theorem for H^s spaces.

Theorem 6.5 Sobolev embedding theorem for H^s
Let $d \in \mathbb{N}$ and $k \in \mathbb{Z}_{\geq 0}$.

(i) *If $s > d/2$, then $H^s(\mathbb{R}^d)$ is continuously embedded in $L^\infty(\mathbb{R}^d)$*

(ii) *If $s > d/2 + k$, then $H^s(\mathbb{R}^d)$ continuously embeds into $C^k(\mathbb{R}^d)$.*

Proof. For (i), it suffices by Fourier inversion (Theorem 4.6) to show that there exists $C = C(d,s) > 0$ such that $||\widehat{f}||_{L^1} \leq C||f||_{H^s}$ for all $f \in H^s(\mathbb{R}^d)$. Observe that, by Cauchy-Schwarz,

$$\begin{aligned} ||\widehat{f}||_{L^1} &= \int_{\mathbb{R}^d} |\widehat{f}(\xi)|^2 (1+|\xi|^2)^{s/2} (1+|\xi|^2)^{-s/2} \, d\xi \\ &= ||f||_{H^s} \left(\int_{\mathbb{R}^d} (1+|\xi|^2)^{-s} \, d\xi \right)^{1/2} \\ &= ||f||_{H^s} \left(C_d \int_0^\infty (1+r^2)^{-s} r^{d-1} \, dr \right)^{1/2} \end{aligned}$$

Since $2s > d$, the integral is finite.

For (ii), note that, for any $\alpha \in \mathbb{Z}^d_{\geq 0}$ with $|\alpha| \leq k$, we have $D^\alpha f \in H^{s-k}(\mathbb{R}^d)$ with

$$||D^\alpha f||_{H^{s-k}(\mathbb{R}^d)} \leq (2\pi)^{|\alpha|} ||f||_{H^s(\mathbb{R}^d)}$$

since $\widehat{D^\alpha f} = (2\pi i)^{|\alpha|}\xi^\alpha \widehat{f}$. Then we have

$$||f||_{C^k} = \sum_{|\alpha|\leq k} ||D^\alpha f||_\infty \leq \tilde{C} \sum_{|\alpha|\leq k} ||D^\alpha f||_{H^{s-k}} \leq C||f||_{H^s}$$

It remains to show that $D^\alpha f$ is continuous for all $|\alpha| \leq k$. To this end, we pick $\varphi \in C_c^\infty(\mathbb{R}^d)$ with $\int \varphi\, dx = 1$ and $\varphi \geq 0$ everywhere. We then set $\varphi_r(x) = r^d\varphi(rx)$. By Proposition 6.3, we see that $\varphi_r \star f \to f$ in $H^s(\mathbb{R}^d)$ as $r \to \infty$. By (i), we deduce that $\varphi_r \star f \to f$ uniformly. Since each $\varphi_r \star f$ is continuous, we deduce that $f \in C(\mathbb{R}^d)$. A similar argument works for $D^\alpha f$. ■

6.3 Trace theorem

In PDE analysis, we often specify boundary conditions that a solution must satisfy. Since weak solutions belonging to a Sobolev space are merely defined a.e. and the boundary of a domain has measure zero, we run into a problem when trying to specify boundary conditions. This motivates the introduction of the so-called *trace operator*, which allows us to do so under weaker regularity assumptions compared to the remedy that the Sobolev embedding provides via the existence of a continuous representative.

Define the half-space $\Omega := \mathbb{R}_{>0} \times \mathbb{R}^{d-1} \subset \mathbb{R}^d$.

Definition Sobolev space $H^s(\Omega)$
For $s \in \mathbb{R}_{\geq 0}$, we define the Sobolev space $H^s(\Omega)$ as

$$H^s(\Omega) := \left\{ f|_\Omega \colon f \in H^s(\mathbb{R}^d) \right\}$$

endowed with the norm

$$||f||_{H^s(\Omega)} = \int \{||g||_{H^s(\mathbb{R}^d)} \colon g \in H^s(\mathbb{R}^d), g|_\Omega = f\}$$

It is a non-trivial fact (which we do not prove in this course) that this norm is equivalent to the $W^{s,2}(\Omega)$ norm.

Theorem 6.6 Trace theorem
Let $s > \frac{1}{2}$. Then there exists a bounded linear operator $T \colon H^s(\Omega) \to H^{s-1/2}(\partial\Omega)$ such that

$$Tf = f|_{\partial\Omega}$$

for every $f \in H^s(\Omega) \cap C(\overline{\Omega})$.

The operator T in the theorem above is called the *trace operator*. You will be asked to prove this result on Example Sheet 4.

6.4 Rellich–Kondrachov theorem

Let $\Omega \subset \mathbb{R}^d$ be an open set. We may regard every $f \in C_c^\infty(\Omega)$ as a function on $\mathbb{R}^d$ by setting $f \equiv 0$ on $\mathbb{R}^d \backslash \Omega$.

Definition Sobolev space $H_0^1(\Omega)$
We define the space $H_0^1(\Omega)$ as

$$H_0^1(\Omega) := \overline{C_c^\infty(\Omega)}^{H^1(\mathbb{R}^d)}$$

endowed with the $H^1(\mathbb{R}^d)$ norm.

Remarks

- Formally, the elements of $H_0^1(\Omega)$ are functions on $\mathbb{R}^d$ but they vanish on $\mathbb{R}^d \backslash \Omega$ so there is no harm in identifying them with their restriction to Ω.
- If Ω is a sufficiently nice domain (e.g. $\Omega = \mathbb{R}_{>0} \times \mathbb{R}^{d-1}$ or a bounded domain with C^1 boundary), then $H_0^1(\Omega)$ is the kernel of the trace operator $H^1(\Omega) \to H^{1/2}(\partial\Omega)$.

Now, the space $H_0^1(\Omega)$ is a closed subspace of $H^1(\mathbb{R}^d)$ so it is a Hilbert space and thus we know that its unit ball $B_{H_0^1}(\Omega)$ is compact in the weak topology. Thus, every sequence has a weakly convergent subsequence. In the analysis of PDEs, it would be useful to have a subsequence that converges strongly. By the following theorem, this is indeed possible in the L^2 norm.

Theorem 6.7 Rellich–Kondrachov theorem
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain. Let $(f_n)_{n\in\mathbb{N}}$ be a bounded sequence in $H_0^1(\Omega)$. Then there exists a subsequence $(f_{n_j})_{j\in\mathbb{N}}$ and $f \in H_0^1(\Omega)$ such that
(i) $f_{n_j} \rightharpoonup f$ *in* $H_0^1(\Omega)$
(ii) $f_{n_j} \to f$ *in* $L^2(\Omega)$

Corollary 6.8
Let $\Omega \subset \mathbb{R}^d$ be a bounded domain. Then $B_{H_0^1}(\Omega)$ is compact in the norm topology of $L^2(\Omega)$.

Proof. By Banach–Alaoglu (Theorem 3.11), there exists a subsequence $(f_{n_j})_{j\in\mathbb{N}}$ such that $f_{n_j} \rightharpoonup f$ for some $f \in H_0^1(\mathbb{R}^d)$. For each j, set $g_j = f_{n_j} - f$. It remains to show that $g_j \to 0$ in $L^2(\Omega)$.

Note that, for every $h_1 \in H^1(\mathbb{R}^d)$,

$$\langle g_n, h_1 \rangle_{H^1} = \int (1 + |\xi|^2) \widehat{g_n} \overline{\widehat{h_1}} \, d\xi \to 0$$

Moreover, for each $h_2 \in L^2(\mathbb{R}^d)$, we have $\widehat{h_2} \in L^2(\mathbb{R}^d)$ and thus

$$\int (1 + |\xi|^2)^{-1} |\widehat{h_2}(\xi)|^2 \, d\xi = \int (1 + |\xi|^2) \left| (1 + |\xi|^2)^{-1} \widehat{h_2}(\xi) \right|^2 \, dx < \infty$$

It follows that $(1 + |\xi|^2)^{-1} \widehat{h_2} = \widehat{h_1}$ for some $h_1 \in H^1(\mathbb{R}^d)$. ∎